

1 Introduction

A rise in internet usage over the past twenty years [1, 2] has created many opportunities for business to grow and profit. With the state of the e-commerce industry, worth 431 648 million USD and projected to reach 563 388 million USD by 2025 in the United States of America [3], the incentive to grab a share of this increasingly popular platform is at an all time high. However, with a new platform that essentially encapsulates the idea of a virtual market, the emergence of recycled retail strategies and new digital persuasive techniques are on the rise. In this new digital environment, it is the friendly and welcoming design of an interface that greets consumers when they first enter an online store. Enthusiastic pop-up ads welcome the consumers and clutter up the interface with the great deals and sales that always seem to be available on e-commerce sites. As the old saying goes, the walls have ears and social media does too. Anything to personalize the user experience, such as encouraging users to buy that lawn mower they mentioned needing a few days back, can be conveniently arranged. Websites are the ones who politely ask for your cookies by pushing a giant “yes” button with the offer of improved experience on the web. Psychological pricing (ex. Product sells for \$4.99 instead of \$5) is only one of many psychological tactics (ex. anchoring, artificial scarcity, decoy effect) employed by online retailers in the hopes of nudging consumers into spending more than they normally would have.

These deceptive tactics are a mixture of old retail practices and new manipulative techniques that have a basis in various fields of research, such as human psychology, behavioral economics, philosophy, ethics, and law to name a few [4, 5]. The evolution of dark pattern from old retail tactics to the adoption of nudging techniques and eventually “growth hacking” illustrates the constantly changing environment that surrounds this field of deceptive online tactics [5]. These tactics are called dark patterns – “tricks used in websites and apps that make you do things that you didn’t mean to,” as stated by Harry Brignull [6]. With more on the rise and a lack of accountability for using these unethical practices, many are calling for more regulation on the usage of digitally deceptive tactics [7, 8, 9, 10].

We add to the vast collection of dark pattern literature by analyzing time spent on an application as a factor that could potentially affect the efficacy of dark patterns on said application. Building on the prior literature that have already analyzed culture, aggressiveness of dark patterns used, and education [8, 11], we focus solely on the expenditure of time (frequent vs non-frequent users) in relation to similar variables analyzed by others such as emotions felt and awareness of dark patterns [8, 23]. We analyzed these factors on users of three e-commerce websites (Amazon, AliExpress, Wish) and three mobile games (Pokémon Go, Subway Surfer, PlayerUnknown’s Battlegrounds Mobile). Ultimately, our purpose is to determine another factor that could potentially affect user’s susceptibility to dark patterns, paving the way for further development in coping or defense techniques that can help protect the population.

We developed an e-commerce and mobile platform survey that was distributed to participants on Amazon Mechanical Turk, a website that provides survey participants for businesses and researchers alike. Survey participants were asked to share their emotions while using their respective platform. They were also primed on dark patterns or game mechanics before being tested on awareness of dark patterns in their e-commerce or mobile application. We utilized the taxonomies developed by Zagal et al. (2013) and Gray et al. (2018) to educate participants on potential dark patterns.

This paper provides a basis for discovering a relationship between the frequency of use and awareness of dark patterns. We aim to seek whether frequent users are more aware of dark patterns in the applications they use compared to non-frequent users.

2 Literature Review

2.1 Taxonomy of Dark Patterns

Dark patterns have been extensively analyzed and thoroughly categorized by the academic community in conjunction with the rapidly growing presence of online activity and internet growth [1, 2]. A diverse selection of taxonomies now exists to aid in the classification of dark patterns, all of which stem from the original works of Harry Brignull who first used the term (dark pattern) in July of 2010 [13]. Brignull later proposed an initial list of eleven different types of dark patterns (confirmshaming was added later) that were shared on darkpatterns.org – a website dedicated to exposing and labelling dark patterns [6].

However, while the initial list was adequately comprehensive at the time of its proposal, Brignull's dark patterns had a few limitations. For example, roach motel, misdirection, and bait and switch were considerably more abstract in concept compared to other identified dark patterns. The varying levels of specificity ranged from vague, as seen in the examples above, to highly specific, such as sneak into basket. Differing levels of specificity also introduced overlap in the conceptualization of individual dark patterns. For example, trick questions can utilize the concept of misdirection to achieve its purpose, and friend spam is an undesirable outcome which a user did not intend to consent to, like the concept of bait and switch. Furthermore, the addition of confirmshaming, which was not part of the original eleven dark patterns, illustrated the need for taxonomy with a hierarchical organization of abstract dark pattern concepts.

Ameliorating upon Brignull's taxonomy, Gray et al. (2018) generated a corpus of dark patterns, extracted from search engines and popular social media platforms, from which they organized these artifacts by five distinct themes: nagging, obstruction, sneaking, interface interference, and forced action [10]. Gray et al. (2018) eliminated the limitations of Brignull's taxonomy by introducing a hierarchy of categories to address the issues of specificity and overlap of concepts. Abstract concepts at the top of the hierarchy allowed for more specific examples to be sorted within the five main domains. For example, roach motel and price comparison prevention were organized under the category of obstruction due to their

overlapping dark pattern concepts (i.e. both created situations that were more difficult than necessary). Other studies have focussed on specific attributes related to dark patterns. For example, Greenberg et al. (2014) provided eight examples of proxemic dark patterns [14] while Bosch et al. (2016) developed a taxonomy based on privacy dark patterns [15]. In 2010, Conti & Sobiesk contributed a taxonomy of ten 'malicious interface techniques' derived from observations in the wild [16]. Alternatively, Mathur et al. (2019) delved into categorizing dark patterns by their characteristics (categorization based on what they do descriptively), rather than organizing by what they are, physically [17].

Studies have also focussed directly on certain platforms of choice. For example, Zagal et al. (2013) developed three dark pattern categories (temporal, monetary, and social capital-based) for dark patterns found specifically in game design [12]. Dark patterns such as playing by appointment provided real-life examples of the difference between what could be considered a dark pattern versus a game mechanic (i.e. not dark a dark pattern). FarmVille – a mobile farming simulator, enabled players to harvest the crops they planted in-game after waiting for a specific amount of time. While the harvesting mechanic in many farming simulators (such as Hay Day) are not necessarily nefarious, what makes FarmVille especially malicious is its withering mechanic which destroys the crop if the player doesn't harvest soon after it's ripe [12].

Lewis (2013), a contributing author of 'Dark Patterns in the Design of Games' also developed a taxonomy for dark patterns specifically in relation to motivational design. Utilizing the same three dark pattern categories (temporal, monetary, and social capital-based), Lewis (2013) described a total of eight dark patterns, six of which overlapped with those mentioned by Zagal et al. (2013). However, Lewis expanded on the ideas from 'Dark Patterns in the Design of Games' and provided a concrete framework to distinguish dark patterns from game mechanics for some of the dark patterns. For example, Lewis (2013) purported

that Dark Pattern: Grind was, “a series of similar tasks a player performs in order to achieve a reward, and that reward does not require skill to achieve. [19]”

Alternatively, Geronimo et al. (2020) studied 240 mobile apps for the pervasiveness of dark patterns on the Android mobile platform, 40 of which were chosen to be analyzed for dark patterns. A comprehensive list of 31 dark patterns was formulated from the 40 apps and each dark pattern was sorted utilizing the taxonomy proposed by Gray et al. (2018) [20, 10].

In contrast to Harry Brignull’s darkpatterns.org website, a reddit users with the handle u/offsky, developed www.darkpatterns.games – a website that catalogues dark patterns found in games [21]. The website derives information from Zagal et al. (2013) in addition to providing a fourth category that is dedicated to purely psychological dark patterns. In total, it provides a comprehensive list of 32 examples of dark patterns found in games.

2.2 Factors Affecting Dark Patterns

Transitioning from classification, recent work in dark pattern literature has looked into the various factors that contribute towards the successful implementation of dark patterns and their efficacy. Prior work by Conti & Sobiesk (2010) analyzed the levels of frustration felt by users who were exposed to dark patterns [16]. However, more recent studies have delved even further.

Geronimo et al. (2020) conducted a study of user perceptions and awareness of dark patterns for mobile applications [20]. The first section of the study was dedicated to the identification and classification of dark patterns from 240 pre-selected apps. The latter part of the study consisted of a survey that analyzed participants' awareness of dark patterns within three (out of five) apps. For each of the three apps, participants were questioned on their ability to spot ‘malicious design’ and were later quizzed on whether the ‘malicious design’ they had discovered aligned with the identified dark patterns from the research team. Researchers utilized a Dark Pattern (DP1) - Dark Pattern (DP2) - No Dark Pattern (NDP) formula for presenting the three apps to measure the survey participant’s initial awareness of dark patterns (DP1), awareness after gaining knowledge of dark patterns (DP2), and awareness of the lack of dark patterns (NDP). The study confirmed that users were more aware when searching for dark patterns after having been primed on the topic [20].

A different study employed a heuristic evaluation to analyze the frequency in occurrence of dark patterns within casual mobile games [22]. Five undergraduate computer science students were chosen as evaluators for four casual mobile games. Utilizing the taxonomy created by Zagal et al. (2013), each dark pattern was analyzed for severity on a ranking scale of 0 (“This dark pattern is not found in the game.”) to 4 (“This dark pattern actively harms not only the playing experience, but also the player’s mentality in approaching the games or ones similar to it (e.g. conditions them to rely on the in-app purchases whenever they face a difficult situation), or even quitting the game entirely.”) [22]. It was noted that pay to skip, grinding, and playing by appointment were the most common tactics found on the four casual mobile games evaluated [22]. However, it was noted that while casual mobile games might employ game mechanics like grinding and playing by appointment, that did not indicate that the use of those game mechanics was inherently dark. It was only when the game provided two options: be forced to partake in the adverse effects of dark patterns or take an action to the benefit of the developers, were these game mechanics to be considered dark [22].

Maier & Harr (2020) performed a study on university-level students, testing their awareness of dark patterns. Participants were given a short PowerPoint on dark patterns before interviewed by qualitative questions. It was noted that manipulation literacy did not provide immunity from dark patterns. In fact, Maier & Harr (2020) suggested that factors like addiction may contribute to people’s acceptance of being manipulated, even when aware [23]. Expanding on the works of Maier & Harr, Colin et al. conducted an experiment on English and Mandarin-speaking participants in a larger scale study. Both groups were

analyzed on how they perceived and responded to dark patterns. Colin et al. noted that participants responded with harsh emotions, such as distress, being upset, hostility, and irritableness, after experiencing dark patterns [8].

This phenomenon of negative emotions in response to dark patterns was noted in a study done by Luguri & Strahilevitz (2021). The mid-section of their three-part study on dark patterns analyzed two groups exposed to mild and aggressive dark patterns. The objective was to convince both groups to enroll in an identity theft protection program. When compared to a control group, Luguri & Strahilevitz (2021) found that the mild group had double the amount enrolled while the aggressive group had quadrupled [11]. However, a more noticeable ‘backlash’ was observed from the aggressive group. Another interesting discovery was the disparity in susceptibility to dark patterns between educated and less educated participants. Less-educated participants were more susceptible to the mild dark patterns compared to their more educated counterparts [11].

2.4 Current Study

While aspects of our paper are reflected in other studies previously done, our research deviates from prior work by relating all data collected to the frequency of use. In the field of study that focuses on studying factors that affect the efficacy of dark patterns, we contribute to this knowledge by sharing how frequent users perceive dark patterns. We utilize current users of the e-commerce platforms and mobile games to collect our data. Finally, the taxonomy systems developed by Gray et al. (2018) and Zagal et al. (2013) are of particular importance to this paper as both are utilized in the formation of the survey and for data analysis.

We also analyze this study in relation to the Persuasion Knowledge Theory (PKM) which suggests that users will have a heightened sense of protection against persuasion attempts if they can identify or are aware that they are being targeted by said attempt [24]. Developed by Friestad & Wright in 1994, the PKM is a lens through which we can potentially attempt to explain whether prior knowledge is an adequate defense against dark patterns. Lewis (2013) also purported a similar idea with manipulation literacy which he proposed that acquiring more manipulation literacy would enable users to make informed decision to protect themselves against the potential threats of deceptive dark patterns [19]. Both proposed the idea that negative feeling could arise against dark patterns or advertising techniques after an adequate amount of persuasion knowledge or manipulation literacy is attained. However, Evans & Park (2015) proposed an alternative view which proposed that there are other ends that could be achieved rather than simply assuming that only negative feelings could arise after gaining knowledge of deceptive advertising techniques [25].

3 Methodology

To achieve our goal of analyzing factors that affect the efficacy of dark patterns, we decided to develop a survey to acquire the necessary data. The survey incorporated a mixed methods design by using a series of multiple choice, Likert scale, slider and long answer questions. We chose three e-commerce platforms (AliExpress, Amazon, Wish) and three mobile games (PUBG Mobile, Pokémon Go, Subway Surfer) to analyze dark patterns and dark game design patterns within the six applications. Official survey participants were collected from Amazon Mechanical Turk and we narrowed the criteria for participants by only allowing those who were currently using the various e-commerce and mobile games. Ultimately, the sought to collect adequate data that could answer the following questions:

1. How does prior knowledge affect the efficacy of dark patterns on e-commerce and mobile game platforms?
2. What emotions are felt by frequent users when utilizing e-commerce and mobile game platforms?
3. How aware are frequent users of manipulative techniques in the applications that they use?

3.1 Survey Sources

Our survey design relies heavily on the taxonomies developed by Gray et al. (2018) and Zagal et al. (2013) [10, 18]. We chose these taxonomies due to their relevance in recent literature [FIND CITATIONS] and their ease of explainability. Because our surveys involved a learning component for the participants, we had to make sure that our videos and written transcripts were concise and engaging. We used all five dark pattern categories from Gray et al. (2018) and six of the dark game design patterns mentioned by Zagal et al. (2013). We chose to omit pre-delivered content from the dark game design patterns taxonomy because the three mobile games we chose to analyze did not require an upfront purchase to play. We also chose to add pay wall, infinite treadmill, and power creep as three extra categories for our mobile application survey as these covered additional ground in manipulative techniques utilized by games.

We also cited Hanlon's razor as a way of briefly introducing the idea of anti-patterns for e-commerce participants [26]. We questioned participants on their experiences with anti-patterns (if they had any) and prompted them to share their own examples. We did not explicitly mention anti-patterns in our mobile application survey. However, we did ask a similar question by prompting mobile participants to share their opinions on what they perceived as 'dark' or not 'dark' within the games they played.

3.2 Survey Design

We issued six final surveys from two unique template surveys which were designed specifically for the e-commerce and mobile applications. Each application (i.e., AliExpress, Amazon, Wish, PUBG Mobile, Pokémon Go, Subway Surfer) had its own survey. Both template surveys had similar sections, but the e-commerce survey only included the dark pattern taxonomy developed by Gray et al. (2018) while the mobile survey only included the three additional categories (pay wall, infinite treadmill, and power creep) and taxonomy provided by Zagal et al. (2013).

The first section of the surveys involved inclusion criteria to weed out the invalid participants. For example, we excluded anyone who had not used one of the six applications within the past month. The mid-parts of the survey involved acquiring data on participants usage of the application (ex. daily, weekly, or monthly), emotions felt, dark patterns noticed, prior awareness of dark patterns, and opinions on dark patterns. To achieve this, we asked users to rate, on a Likert scale, how much they agreed with statements such as, "I have felt disappointed while using [application]," or "I have noticed nagging while using [application]." We also asked two open ended questions (the final open-ended question differed depending on the application's platform):

Q1: Could you share the most recent time you felt manipulated/tricked?

Q2.1 (e-commerce): Could you share your experience with anti-patterns?

Q2.2 (mobile): Could you elaborate why the considered game mechanics are dark game design patterns?

We utilize the term, game mechanics, as a neutral proxy for the term dark game design pattern. The purpose of this is to not perpetuate to the reader the idea that we should automatically consider these categories as 'dark'. As stated by Deterding et al. (2020), blindly stating such categories as 'dark' before we have even evaluated them with an ethical framework is counterproductive in the grand scheme of analyzing ethical game design [29]. The mobile survey design took this into account, and we made sure to never mentioned the term, 'dark game design pattern' until the open-ended Q2.2. As such, we will also refer to these categories as game mechanics in the latter parts of this paper.

The last section of the survey was dedicated to demographic data such as age, gender, country of origin, education, wage, and profession. We also included a validation code at the end of our survey so participants could submit it to Amazon Mechanical Turk to receive payment for their participation.

We also performed a preview test of the e-commerce and mobile template surveys. We shared the surveys on various forums and social media sites. We received 143 responses and adjusted our survey templates based off the feedback given.

3.2 Survey Distribution

Our surveys were published on Amazon Mechanical Turk (AMT). We decided to cap participants at 150 for each survey and we provided a \$0.10 rate for completion. Data was exported from Qualtrics to excel for further analysis.

3.3 Survey Demographics and Statistics

For data analysis, we kept the survey responses separate due to the variation in e-commerce websites and mobile games. This variation affected how participants responded to similar questions on different applications.

We received 836 responses in total, split between six different surveys (AliExpress: n = 134, Amazon: n = 119, Wish: n = 140, Pokémon Go: n = 149, PUBG Mobile: n = 143, Subway Surfer: n = 151). Collectively, we had 600 completed responses that passed the inclusion criteria (AliExpress: n = 98, Amazon: n = 100, Wish = 100, PUBG Mobile: n = 102, Pokémon Go: n = 100, Subway Surfer: n = 100). Within these valid responses, the majority of the participants were male (n = 435, 72.5%), followed by female (n = 161, 26.83%), Other (n = 0, 0%), and four participant did not choose to disclose their gender (picked option 'Prefer not to answer'). The United States of America (n = 164, 27.33%), India (n = 244, 40.67%), and Brazil (n = 94, 15.67%) had the most participants present within the study. All other countries were grouped into an 'Other' category (n = 98, 16.33%) as their participant numbers were considerably lower than the top three when analyzed non-collectively within the six surveys. In total, the distribution of education was reported: < Highschool (n = 3, 0.5%), Highschool (n = 49, 8.17%), College Diploma (n = 74, 12.33%), Undergraduate Degree (n = 329, 54.83%), MBA (n = 22, 3.67%), Masters (n = 117, 19.5%), PhD (n = 4, 0.67%), and Other (n = 2, 0.33%). Age distribution was reported: < 20 (n = 10, 1.67%), 20-25 (n = 216, 36%), 26-35 (n = 274, 45.67%), 36-45 (n = 61, 10.17%), 46-55 (n = 26, 4.33%), 56-65 (n = 12, 2%), 66> (n = 0, 0%), and one participant did not disclose their age.

3.4 Data Analysis

We provide a descriptive analysis of our data in Section 4. Within this section, we analyze user's awareness of dark patterns prior to the survey and provide visuals to accompany our findings. Specifically, we analyze user's general awareness¹ of dark patterns and game mechanics and compare that between the six applications. We also analyze awareness and participants felt manipulation. The purpose of this is to determine if there is a trend between having prior awareness of dark patterns or game mechanics in an application and feeling manipulated by said application. Later, we analyze awareness and emotions felt to determine the differences in emotions for aware and unaware users. We also allowed users to elaborate on which dark patterns or game mechanics they had been aware of which allows us to report the most and least familiar dark patterns/game mechanics. Awareness was also analyzed in relation to gender, age, country, and education. We also provide examples of participants responses to the open-ended questions, Q1, Q2.1, and Q2.2.

Footnote[1]: When mentioning awareness, we do so in reference to the survey which asked participants if they had been aware of the dark patterns or game mechanics, we had mentioned in the survey they were participating in (a "yes" or "no" question). When mentioning 'aware participants', we mean participants

who indicated that they had prior knowledge of dark patterns or game mechanics before the survey. ‘Unaware participants’ or ‘non-aware participants’ indicates those who had no prior knowledge of dark patterns or game mechanics.

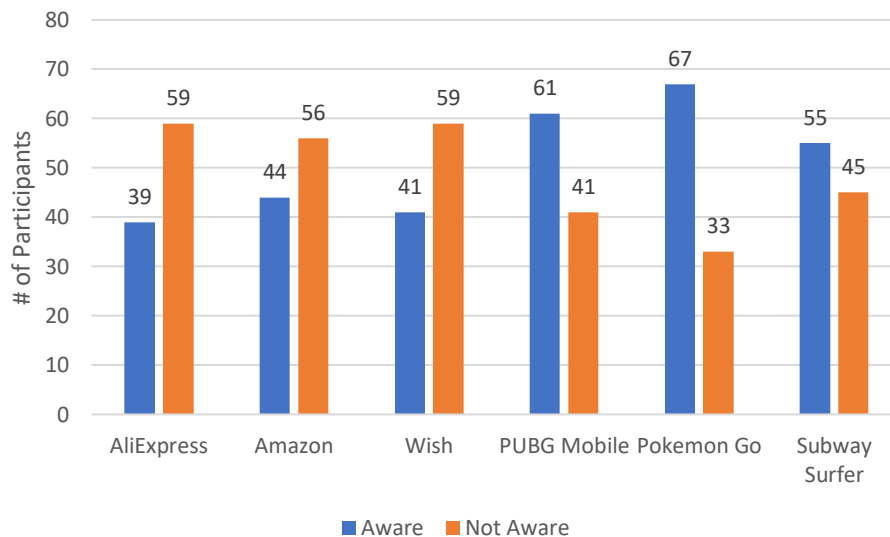
In Section 5, we provide a data analysis of frequent and non-frequent users in relation to emotions felt and dark patterns noticed. For this, we utilized a t-test to check for any statistically significant means between frequency of use, emotions felt, and dark patterns noticed. We also performed a correlation test between emotions felt and dark patterns noticed to see if there were any significant correlations.

5 User’s Awareness of Dark Patterns Prior to Survey

This section details the data and results of participants responses in relation to awareness of dark patterns and game mechanics. After participants had answered questions about noticed dark patterns or game mechanics, we asked them to share which dark patterns or game mechanics they had previously known before taking the survey. We descriptively analyze how awareness and non-awareness are affected by demographic factors such as gender, country of origin, and education. We also describe how awareness and non-awareness affect emotions felt and dark patterns/game mechanics noticed.

5.1 Awareness Prior to Survey

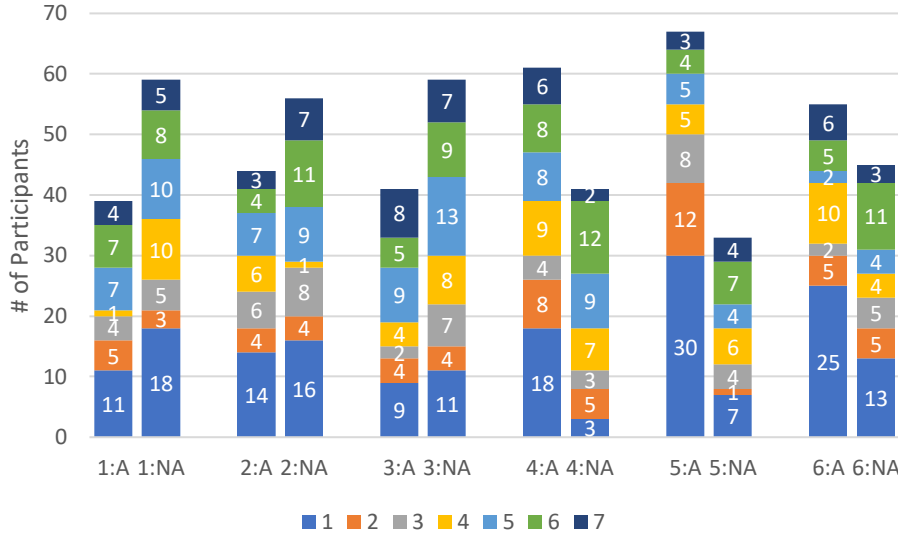
Table #. Participants aware and not aware of dark patterns/game mechanics prior to survey.



After participants had answered questions about noticed dark patterns or game mechanics, we asked them to share whether they had known about these concepts prior to taking the survey. Participants who stated that they had been aware were also prompted to share which specific dark patterns/game mechanics they had been aware of beforehand. The cumulative total of all responses indicated that 51.17% of participants (aware, $n = 307$) had been aware prior to taking the survey whereas 48.83% (not aware, $n = 293$) indicated that they had not been aware. However, this even split between all survey participants was not evenly distributed amongst the six applications we analyzed. The e-commerce platforms (AliExpress: 39.8% aware, Amazon: 44% aware, Wish 41% aware) consistently had fewer participants that were aware of dark patterns prior to taking the survey compared to the mobile platforms (PUBG Mobile: 59.8% aware, Pokémon Go: 67%, Subway Surfer: 55%) who consistently had more participants that were aware of game mechanics.

5.2 Awareness and Manipulation Felt

Table #. Likert scale results for manipulation felt by aware and non aware participants.



Note: 1. AliExpress, 2. Amazon, 3. Wish, 4. PUBG Mobile, 5. Pokémon Go, 6. Subway Surfer. A is aware and NA is not aware.

Tips for analyzing chart: This graph represents the results for the question: “I have felt manipulated by [application].” The legend indicates the values of the Likert scale (1: Strongly Disagree with statement – 7: Strongly Agree with statement). When looking at a bar column, the stacked bars indicate the number of participants who chose a specific Likert value for that application. For example, the orange value for the stacked bars in section 5:A read ‘12’. That indicates that 12 people chose the Likert value 2, indicating that they ‘disagree’ with the statement, “I have felt manipulated by [application].”

We asked participants to share whether they had felt manipulated by the application they used. On a scale from 1 to 7 (Strongly Disagree – Strongly Agree), users rated how much they agreed with the following statement, “I have felt manipulated on [application].” Participant’s felt manipulation was relatively similar for e-commerce platforms. 51.28% (n = 20), 54.55% (n = 24), and 36.59% (n = 15) of aware participants for AliExpress, Amazon, and Wish, respectively, chose values between 1 and 3 (Strongly Disagree – Disagree). 44.07% (n = 26), 50% (n = 28), and 37.29% (n = 11) of non-aware participants for AliExpress, Amazon, and Wish, respectively, chose values between 1 and 3 (Strongly Disagree – Disagree). The percentage of aware and non-aware participants on e-commerce applications who chose to disagree, to some extent, with the felt manipulated statement mirrors each other relatively closely. On the other hand, participant’s felt manipulation for the mobile games differed slightly. 49.18% (n = 30), 74.63% (n = 50), and 58.18% (n = 32) of aware participants for PUBG Mobile, Pokémon Go, and Subway Surfer, respectively, chose values between 1 and 3 (Strongly Disagree – Disagree). 26.83% (n = 11), 36.36% (n = 12), and 51.11% (n = 23) of non-aware participants for PUBG Mobile, Pokémon Go, and Subway Surfer, respectively, chose values between 1 and 3 (Strongly Disagree – Disagree). Aware participants of the mobile applications seemed to have a higher tendency for rating that they had not felt manipulated compared to their unaware counterparts.

On the flip side, analyzing the number of participants who chose values between 5 and 7 (Agree – Strongly Agree) also confirmed this trend. 36.07% (n = 22), 17.91% (n = 12), and 23.64% (n = 13) of aware participants for PUBG Mobile, Pokémon Go, and Subway Surfer, respectively, chose values between 5 and 7 (Agree – Strongly Agree). 56.1% (n = 23), 45.45% (n = 15), 40% (n = 18) of non-aware participants for PUBG Mobile, Pokémon Go, and Subway Surfer, respectively, chose values between 5 and 7 (Agree – Strongly Agree). The percentage of participants who were aware and felt manipulated is considerably smaller in comparison with the percentage of participants who were not aware.

5.3 Awareness and Emotions Felt

Participants were given a series of statements similar to, “I have felt [emotion] in [application],” and were able to rank these on a scale of 1 to 7 (Strongly Disagree – Strongly Agree). Results indicated that aware and non-aware participants for AliExpress agreed (values picked between 5 – 7) that they felt positive emotions like delight (aware: n = 24, non-aware: n = 37), excitement (aware: n = 28, non-aware: n = 38), happiness (aware: n = 31, non-aware: n = 40), and satisfaction (aware: n = 30, non-aware: n = 42) more often than negative ones. To contrast, very few people agreed that they felt disappointed (aware: n = 11, non-aware: n = 13), pressured (aware: n = 11, non-aware: n = 14), stressed (aware: n = 13, non-aware: n = 10), or irritable (aware: n = 11, non-aware: n = 10) while using AliExpress. This same trend can be visualized again for Amazon and is illustrated utilizing Tables #. All other applications showed similar trends to this.

Table #. Felt emotions for aware participants on Amazon.

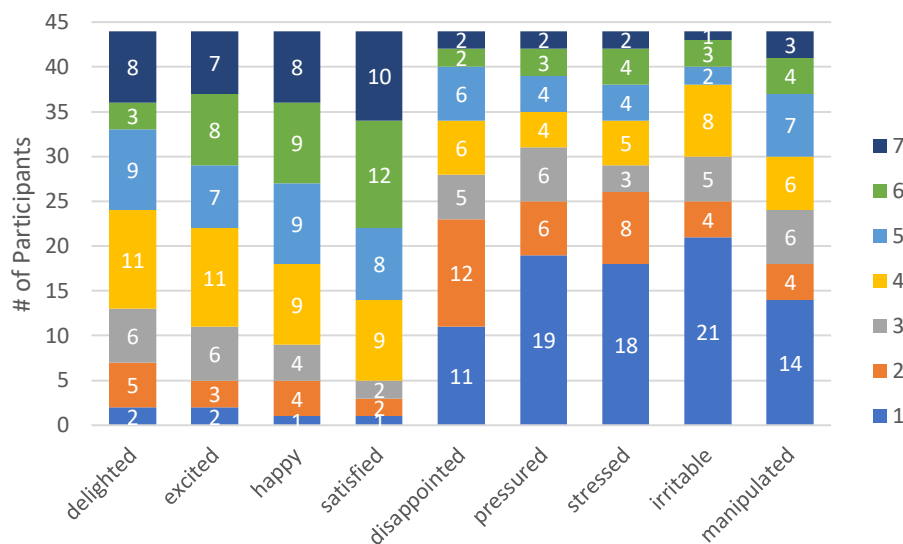
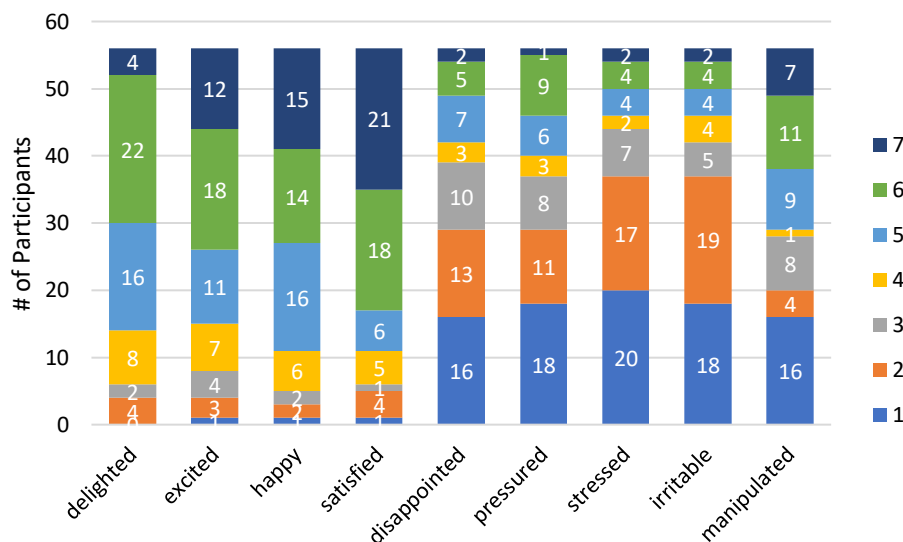
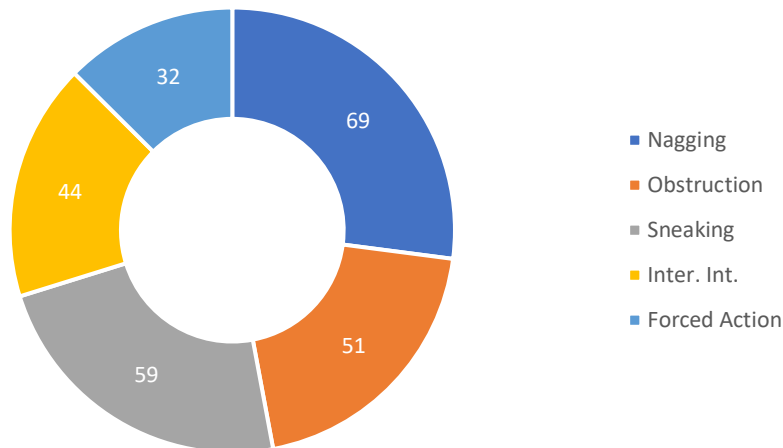


Table #. Felt emotions for non-aware participants on Amazon.



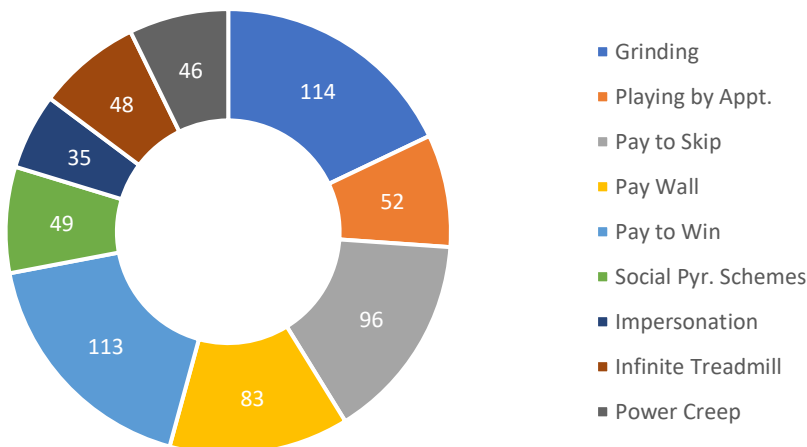
5.4 Most and Least Well-Known Dark Pattern and Game Mechanic

Table #. Dark Patterns Aware of Prior to Survey.



Participants who had stated that they had been aware of dark patterns prior to the survey were also given the chance to share which ones. In total, 124 e-commerce participants indicated that they had been aware, prior to taking the survey. Each of those participants were allowed to indicate one or more dark patterns that they had been familiar with. We noted that nagging ($n = 69$, 55.65%) was the most familiar e-commerce dark pattern concept that people had been aware of prior to taking the survey. Forced action ($n = 32$, 25.81%) was a dark pattern concept that people were least familiar with. Sneaking ($n = 59$, 47.58%) was a close second in familiarity followed by obstruction ($n = 51$, 41.13%) and interface interference ($n = 44$, 35.48%).

Table #. Game Mechanics Aware of Prior to Survey.



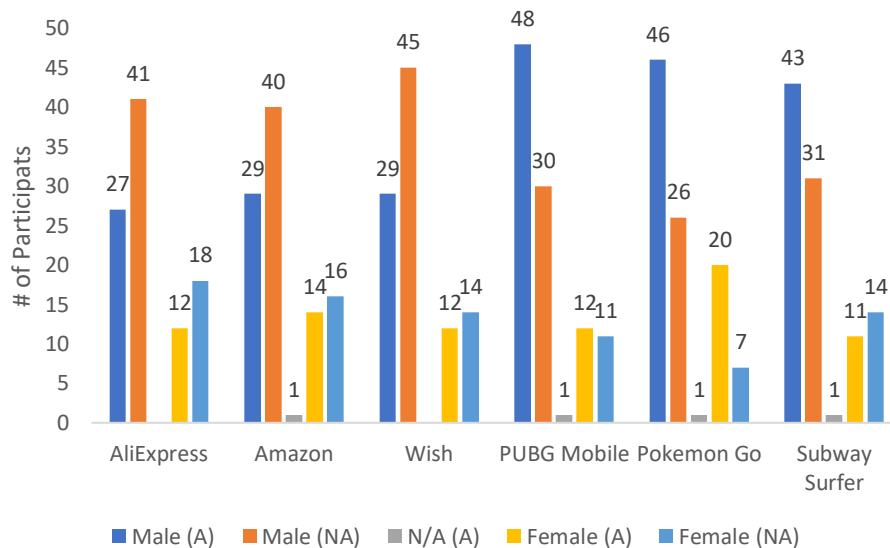
Participants were asked if they had been aware of the mentioned game mechanics prior to the survey and were also asked to elaborate on which ones. In total, 183 mobile game participants indicated that they had been aware. We noted that grinding ($n = 114$, 62.29%) and pay to win ($n = 113$, 61.75%) were the most familiar game mechanics for participants. Pay to skip ($n = 96$, 52.46%) and pay wall ($n = 83$, 45.36%) followed indicating that monetary mechanics were the most familiar to participants. Playing by appointment ($n = 52$, 28.42%) was followed closely by social pyramid schemes ($n = 49$, 26.78%), infinite treadmill ($n = 48$, 26.23%), and impersonation ($n = 46$, 25.14%). Power creep ($n = 35$, 19.13%) was the least familiar concept.

5.5 Awareness and Demographics

We gave participants the option to share their demographic data at the end of the survey. These questions allowed them to share their gender, age, country, and education. In this section, we will see if there are any differences in those categories with respect to awareness.

5.5.1 Gender

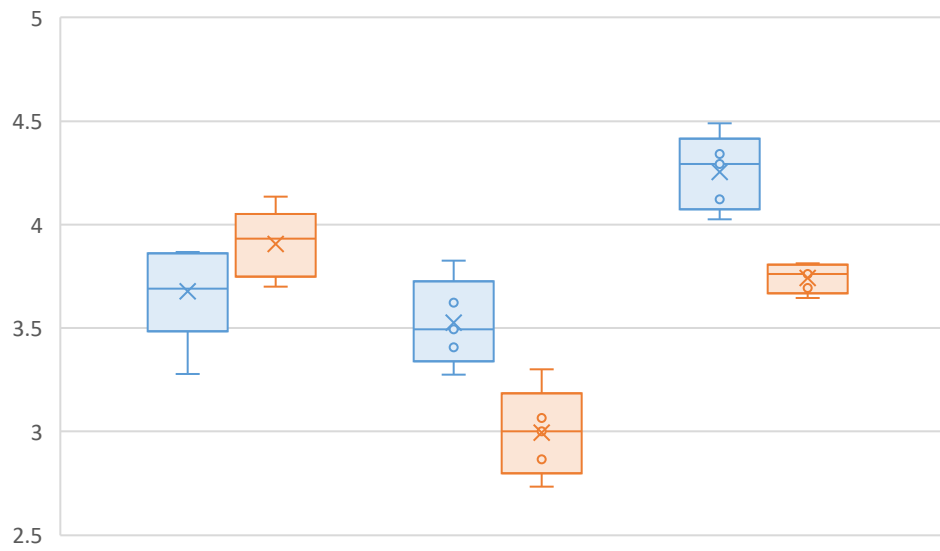
Table #. Awareness in male and female participants.



Note: A indicates aware, NA indicates not aware, N/A indicates those who did not disclose their gender.

As seen in Section 5.1, Table # mirrors the trends seen by Table # (in Section 5.1). The e-commerce platforms consistently had a percentage of around 40.31% (averaging AliExpress: 39.71% aware, Amazon: 42.03% aware, Wish: 39.19% aware) of male participants having been aware, prior to the survey whereas the female participants averaged around a 44.27% (averaging AliExpress: 40%, Amazon: 46.67%, Wish: 46.15%) of participants who were aware. The mobile platforms consistently had a percentage of around 61.02% (averaging PUBG Mobile: 61.54%, Pokémon Go: 63.39%, Subway Surfer: 58.12%) for male participants who were aware. However, this trend was not as consistent for female participants who averaged around 56.75% (averaging PUBG Mobile: 52.17%, Pokémon Go: 74.07%, Subway Surfer: 44%). There were slight differences between genders, in terms of awareness, with female participants having a higher percentage of those aware on the e-commerce platforms and male participants having a higher percentage of those aware on the mobile platforms.

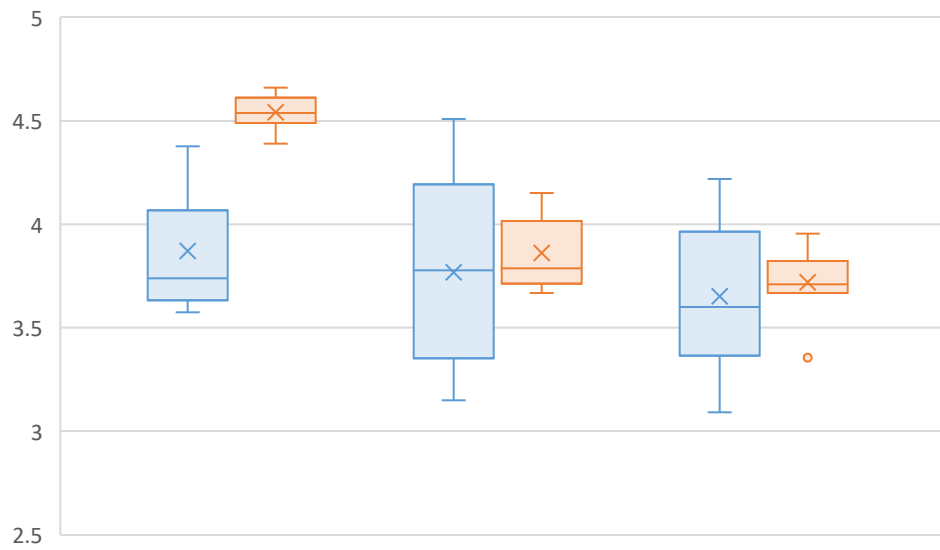
Table #. Mean values of dark patterns noticed in e-commerce platforms in the following order: AliExpress, Amazon, Wish, for male (blue) and female (red) participants.



Note: Blue indicates males, red indicates females. From left to right (grouped by one blue and red box and whisker plot): AliExpress, Amazon, and Wish.

Despite this, Table # illustrates that on average, male participants noticed more dark patterns on Amazon and Wish while overlapping slightly with female participants in dark patterns noticed for AliExpress. The table shows the means of dark patterns noticed, including the max and minimum mean values. In total, Amazon is illustrated to have either less dark patterns or better covert dark patterns compared with AliExpress and Wish.

Table #. Mean values of dark patterns noticed in mobile platforms in the following order: PUBG Mobile, Pokémon Go, Subway Surfer, for male (blue) and female (red) participants.

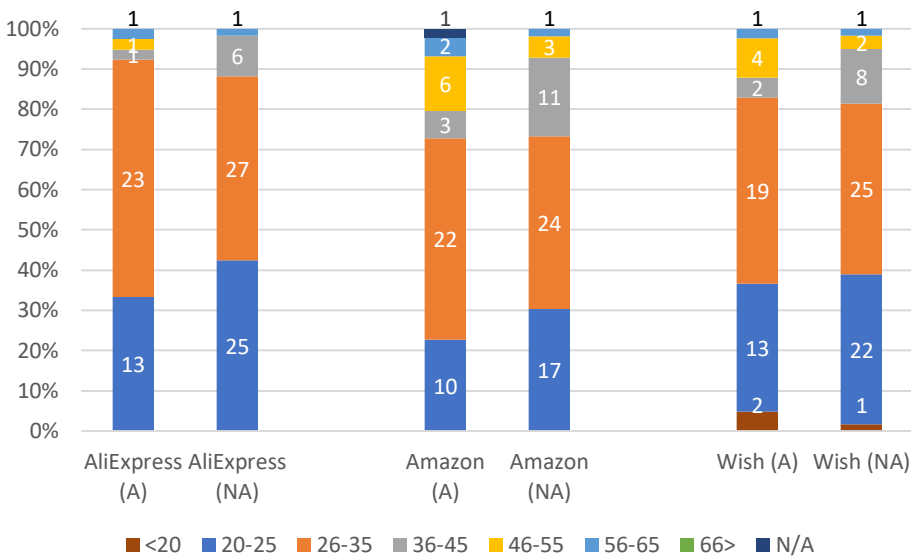


Note: Blue indicates males, red indicates females. From left to right (grouped by one blue and red box and whisker plot): PUBG Mobile, Pokémon Go, and Subway Surfer.

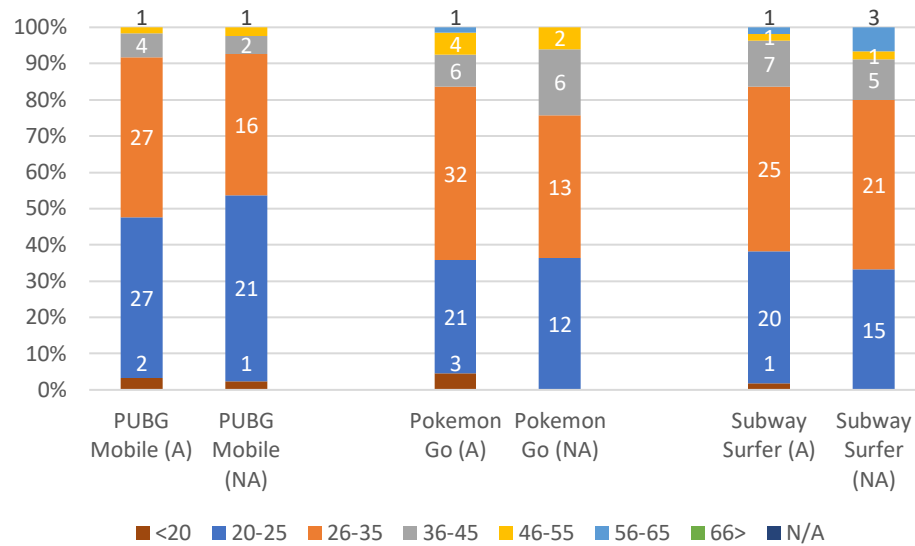
For the mobile platforms, the means are relatively stable for male and female participants from Pokémon Go and Subway Surfer. However, females for PUBG Mobile rated considerably higher for noticed game mechanics compared to the male participants. This is in spite of the fact that male participants of PUBG were noted to have been aware of game mechanics compared to the female participants (male: 61.54% aware, female: 52.17% aware)

5.5.2 Age

Table #. Percentage of aware (A) and non-aware (NA) participants divided by age.

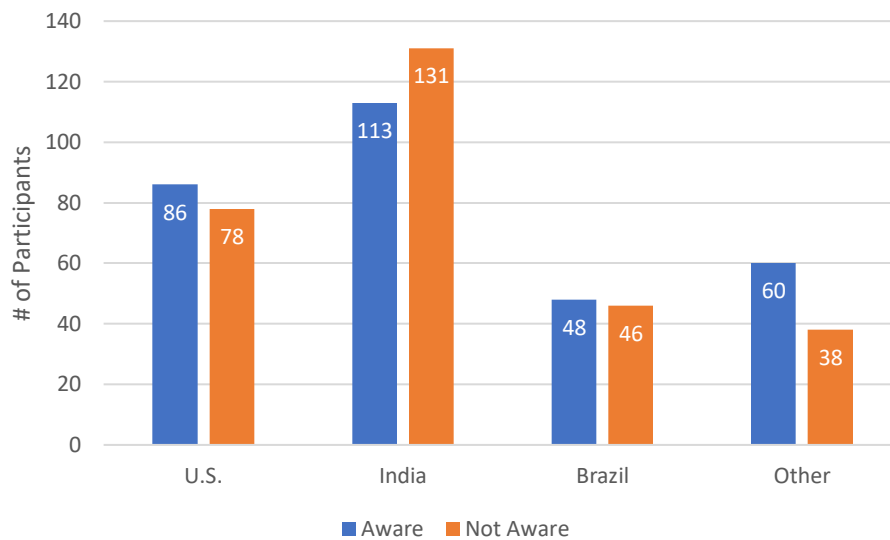


It was noted that the majority of e-commerce participants who were aware and non-aware fell under the age of 35. This was further split between participants who were between the ages of 20 – 25 and participants who were 26 – 35 years of age. Of the 100 participants who indicated that they were 20 – 25 years of age, 36% (n = 36) of them indicated that they had been aware, prior to the survey. 140 participants fell between the ages of 26 – 35. 45.71% (n = 64) of these participants indicated that they had been aware. The remaining 55 participants were distributed amongst other age group categories and 40% (n = 22) indicated that they had been aware prior to the survey.



5.5.3 Country

Table #. Number of aware and unaware participants divided by country of origin.



The majority of our survey participants came from three main countries: United States of America, India, and Brazil. As such, we noted participants from these three countries and grouped all the rest into the “Other” category. It is noted that there was no considerable difference in awareness between the top three countries and we could not compare the other countries due to their low survey participant population. A total of 164 survey participant were from the United States of America with around 52.44% (n = 86) indicating that they were aware. India had the largest representation of population throughout our six survey with 244 participants, 46.31% (n = 113) indicating that they were aware. Participant from Brazil reached 94 with 51.06% indicating they had been aware.

Table #. Number of aware and unaware participants divided by country of origin for e-commerce.

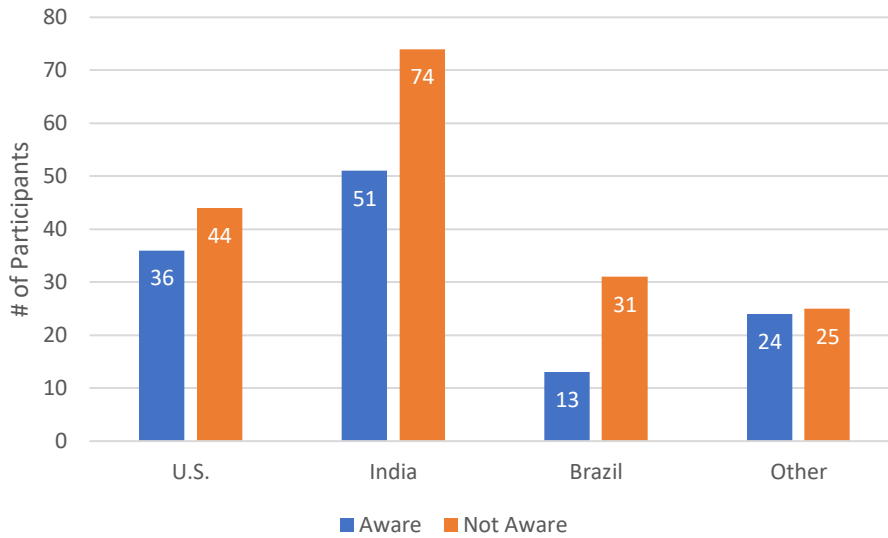
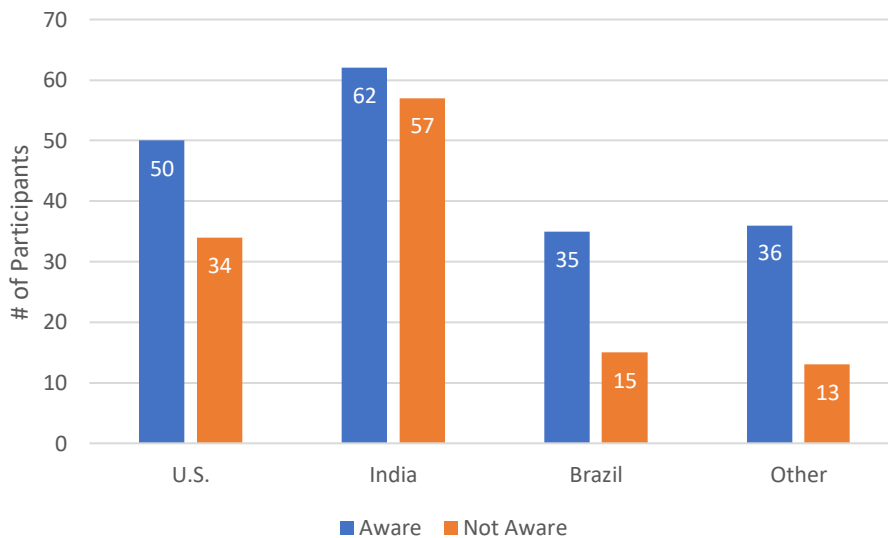


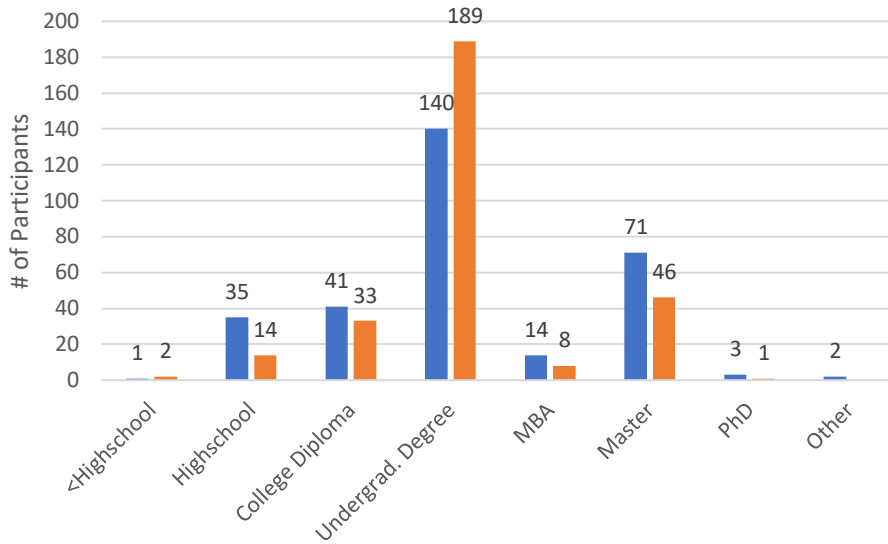
Table # and Table # represent the individual populations from each country for e-commerce and mobile platforms specifically. As seen before in Section 5.1, the percentage of aware participants was consistently lower for e-commerce compared to the mobile platforms. For e-commerce, the U.S. had the highest percentage of aware participants (45%, $n = 36$) followed India (40.8%, $n = 51$) and Brazil (29.54%, $n = 13$). For mobile platforms, Brazil had the highest percentage of aware participants (70%, $n = 35$) followed by the U.S. (59.52%, $n = 50$) and India (52.1%, $n = 62$).

Table #. Number of aware and unaware participants divided by country of origin for mobile.



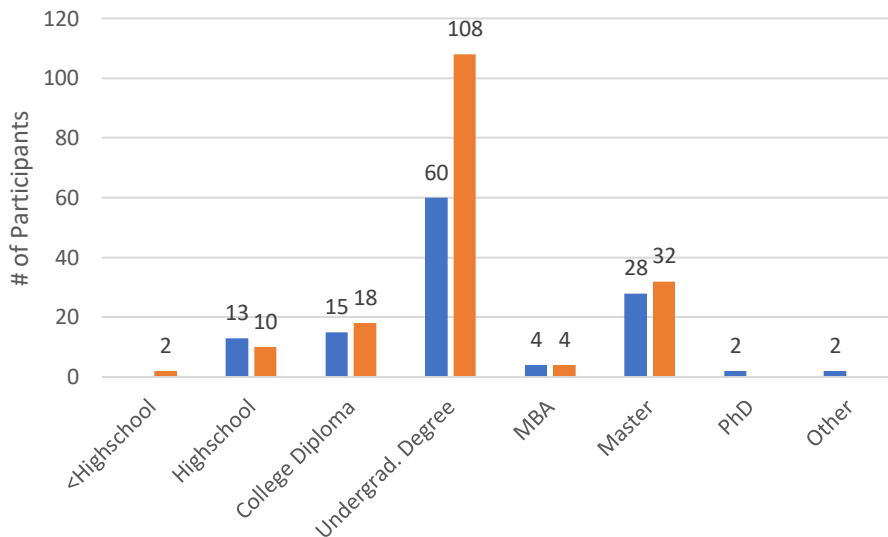
5.5.4 Education

Table #. Number of aware and non-aware participants divided by education.



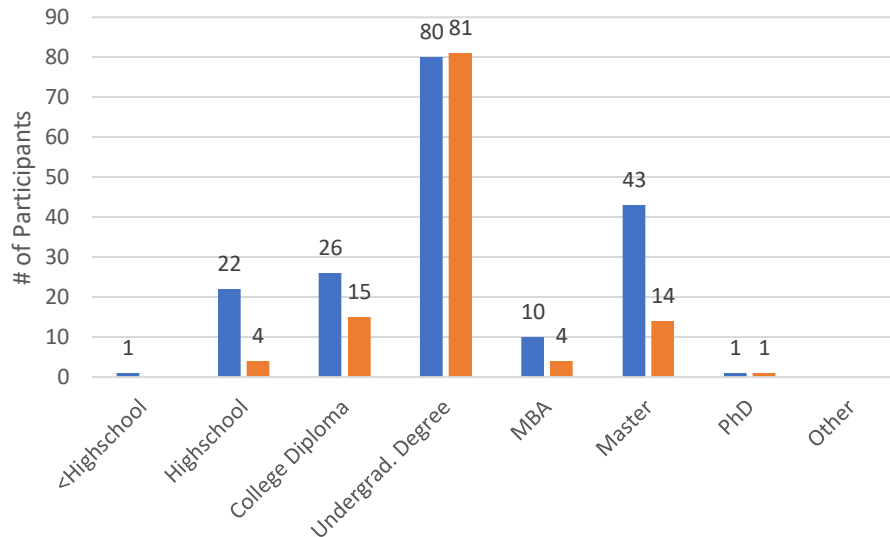
The majority of our survey participants had undergraduate degrees (54.83%, n = 329). 19.5% (n = 117) had Masters, 12.33% (n = 74) had college diplomas, 8.17% (n = 49) had high school diplomas, 3.67% (n = 22) had MBA's, 0.67% (n = 4) had PhD's, 0.5% (n = 3) had did not have high school diplomas, and 0.33% chose the option of 'Other'.

Table #. Number of aware and non-aware participants divided by education for e-commerce.



Our results showed that participants who received their high school diploma, MBA, or Masters had an awareness rate of 56.52% (n = 13), 50% (n = 4), and 46.67% (n = 28). While college diploma (45.45% aware, n = 15) and undergraduate degree (35.71%, n = 60) participants had slightly lower awareness rates, due to the differences in numbers for each category, we could not suggest anything more than mere observations.

Table #. Number of aware and non-aware participants divided by education for mobile platform.



The results for the mobile platforms illustrated that high school diploma participants had highest percentage (84.61%, $n = 22$) of those who were aware of game mechanics. This was followed by participants who held a Masters (75.44%, $n = 43$), college diploma (63.41%, $n = 26$), and MBA (71.43%, $n = 10$). Interestingly enough, undergraduate degrees had an almost even split with 49.69% ($n = 80$) participants claiming to be aware.

6 Qualitative Responses

In this section we provide an analysis of the qualitative data we received from participants who answered the long answer questions in our surveys. We seek to find what users think is manipulative in the applications they use by analyzing their responses. We will provide themes for the e-commerce responses and a case analysis of the Pokémon Go survey responses.

6.1 E-commerce Applications

6.1.1 “Share a recent time you felt manipulated/tricked?”

Our e-commerce participants provided a total of 25 responses for the above question. Within those 25 responses, split amongst the three e-commerce platforms, similar ideas arose. Approximately eight responses talked about buying products that were advertised as one thing but were actually something entirely different. While e-commerce platforms can’t always be blamed for the appearance of fake sellers on their site, it does raise the question of how much accountability can we place on e-commerce sellers to protect shoppers? As one participant stated:

“I wanted to purchase a quality made, name brand waiter's corkscrew and it appeared to be authentic. I started reading the reviews and several of them complained that the product was counterfeit. I was really upset because I feel like Amazon should be responsible for weeding out counterfeit items. If customers have reported that these [counterfeit] items are sold, Amazon shouldn't allow the seller on their platform.”

Six responses mentioned search results and offers. Participants noted that search results for items on Google usually returned with Amazon sellers. Participants expressed their frustrations in seeing Amazon items, “99.9% of the time,” that tended to be out of stock, “75% of the time.”

6.1.2 Anti-Patterns

We asked users to share if they felt they had experienced anti-patterns before and only 5 responses were recorded in total, all of which mentioned ads. One participant noted the frustration in being shown a false ad which redirects the user to news sites that prompt users to subscribe. Another user mentioned having to use ad-blocker to reduce clutter when visiting sites and two others noticed how ads were personalized for them specifically:

“If I liked a post or an ad, it will show up again on my feeds most of the times.”

“It knows my interest [in] products and targets me again with their pop-up ads or [refer] items you may like.”

6.2 Pokémon Go Case Analysis

6.2.1 “Share a recent time you felt manipulated/tricked?”

In total, we had 69 responses for this question. After reading through all the responses, it became clear that there were two distinct themes and a few single responses that outlined examples of how participants had been tricked/manipulated. Disappointing in-game events were mentioned 25 times with the Kanto Tour Pass (i.e., similar to battle/seasons passes in other mobile games) being mentioned 14 times. As a participant put it:

“Pokémon Go Kanto Tour was misleadingly promoted as one that would have shiny rates. Since they don’t disclose rates, they should be more explanatory in their announcements. Very deceitful!”

These shiny rates (shiny – a rare form of a Pokémon that has a unique appearance compared to its normal counterpart, shiny rates – chance of catching a shiny) lead to many participants feeling manipulated since vital information (shiny rates) was withheld from them, inhibiting them from making an informed decision. As another participant mentioned:

“The Snorlax shiny rates on the Go Tour. I wasted 2 hours and many passes trying to get a shiny Snorlax. Felt totally tricked after learning that it was a full odds shiny. It's unprecedented that they release a rare [Poke]mon such as Snorlax with full odds. I was angry and sad at the same time. I don't like feeling like a chump.”

As another point of contention, participants mentioned Pokémon Go’s egg hatching feature 13 times. The egg hatching feature allows users to incubate eggs by walking certain distances (2km, 5km, 7km, 10km, and 12km). However, since Pokémon Go only provides one free incubator (with unlimited uses), users must earn or buy additional incubators to hatch more than one egg simultaneously. Despite this, additional incubators do not guarantee that a user will hatch the Pokémon they want.

“Low hatch rates of some Pokémon that [are] highly advertised in an event that lasts only a few days, requiring the use of many paid incubators to even have a reasonable chance of getting it.”

Other features mentioned that made participants feel manipulated included Remote Raid Passes – a product that can be bought and used to join in gym raids (a Pokémon boss battle) with other users. When a user uses a Remote Raid Pass, they have a chance to catch the Pokémon boss that they battle. Users disliked the restriction on carrying only three Remote Raid Passes at maximum while other users expressed their frustration in remote raids:

“I did buy one remote raid pass and not able to catch the legendary which is annoying since real money was involved but that’s just a personal vent/regret.”

6.2.2 “Do you consider these game mechanics as dark game design patterns?”

We received 50 responses for this question, many of which provided benefits and drawbacks to the various uses of game mechanics within Pokémon Go. As many participants stated ($n = 12$), game mechanics like grinding and infinite treadmill give Pokémon Go purpose and provide its users with a sense of accomplishment. As one user stated:

“Grinding gives the game meaning IMO [in my opinion]. If somebody sees my level 40 shadow MM metagross, they know I put some work into it, so I think that each game needs a little bit of grinding to feel truly satisfying. However, none of this applies to the new XL candy system, my answers about grinding were without the consideration of the XL system, because we are fairly early in its implementation development. If it were to stay the same, move my answers about grinding to 6s or 7s, that system is horrible.”

However, game mechanics that are used for engagement and good purposes can also simultaneously be utilized for potentially frustrating and unfair gameplay. As the participant mentioned above, changes to Pokémon Go’s gameplay also introduced changes to their opinion on whether grinding could be considered ‘dark’. However, this participant also showed lenience in its rating of grinding by providing a certain amount of time to see whether the XL candy system would be changed, illustrating that immediate changes may not immediately change a participant’s opinion on what constitutes as ‘dark’ in games.

Additionally, 22 participants mentioned topics related to real-world currency and monetary dark game design patterns. Most noted that these monetary mechanics were unavoidable and ultimately put free-to-play players at a disadvantage. As one participant mentioned:

“Pay to win/progress always seems negative to me as a mostly free to play type player. The continuation of paywalled new mons is frustrating, but the idea of pay to win has always been part of the game; you buy items such as lucky eggs and you level faster, star pieces and you have more powering up capabilities, not even starting on raid passes.”

Perpetuating the idea mentioned by Zagal et al. (2013) that game mechanics seem dark when they adversely affect a player’s playstyle, this participant illustrates the perceived unfairness of the advantages garnered by pay-to-win players. However, another participant proposed an alternative view of monetary mechanics:

“I also don't think the pay wall for events are bad because Pokémon Go is free. They have to make money somehow. Are we gonna classify games that cannot be played for free as "dark"?”

One user mentioned an interesting monetary topic that was not covered in our survey. In relation to certain gameplay elements of Pokémon Go, this participant argued for the existence of addictive mechanics like gambling within the game:

“One item not mentioned here is gambling. Thinking mostly in terms of incubators for eggs, where you don't know what you will get. Or the ever-popular new shiny being either in raids or eggs, and "if you are lucky" with no published odds. These seem geared towards generating addiction, and I would consider very dark. Although when players will see their kidney for a shiny, id the problem them or the game (not that it should be exploited).”

As demonstrated by previous participant in Section 6.2.1 who expressed regret from buying a Remote Raid Pass, users are essentially spending real-world currency to gain only a chance at attaining a reward.

Mentions of social topics like impersonation and social pyramid schemes were reported 10 times. However, many participants elaborated that they either did not see any existence of social mechanics in Pokémon Go, or they argued against any perceived darkness within the mechanics themselves. As one participant put it nicely:

“Really the social stuff is the stuff that raises my eyebrows the most? The game has this very awkward mix of forcing you to be social but doing absolutely nothing to facilitate that. They offload all the burden and responsibility of the risks of talking to strangers, toxicity of other players, general social awkwardness, etc, onto the players, but reap all the rewards of social obligation. They make trading almost impossible to do unless you become their best salesman and harangue all your friends into becoming regular players. I wouldn't mind the social stuff at all if they actually helped you to make friends, a lot of us lonely gamers could use that, but they don't want to touch actual social contact with a 20 foot pole, because moderating people is actual work and responsibility, and social stuff can go sour fast without moderation. But what that means is we still don't have moderation, and we don't have any scaffolding, structure, or support at ALL. Yet we're expected to go guilt-trip our friends into playing more and have social ties keeping us in the game. It's [like] they want to have their cake and eat it too. This is the one thing they do that actually feels nefarious to me. The rest of the game is honestly a delight and downright mild and harmless by mobile game ethical standards.”

Summed up adequately, while social mechanics can be quite nefarious when implemented with the intent to maliciously use social pressure, social mechanics without such pressure can be considered quite harmless.

An additionally interesting note was the mention of FOMO – Fear Of Missing Out, by a few participants who noted this as an additional game mechanic that was not covered in our survey. Participants linked FOMO to the game mechanic, playing by appointment, indicating that those two went hand-in-hand. Participants noted that players adhere to the game's set schedule (i.e. playing by appointment) due to the fear of missing out on limited time events and items.

4 Results

In this section, we will detail the primary results of our main research questions (“What emotions are felt by frequent users when utilizing e-commerce and mobile games?”, “How aware are frequent users of dark patterns/game mechanics in the applications that they use?”, and “How does emotions felt relate to dark patterns noticed?”). Each subsection is dedicated to one of the six applications we analyzed in this study and we will summarize any statistically significant or insignificant results that we acquired from our statistical tests. A summary of the results is provided at the end (see Section #, Table #).

4.1 AliExpress Results

4.1.1 Felt Emotions. There were no significant differences in the results for delight felt by frequent (M=4.7, SD=1.6) and non-frequent (M=4.8, SD=1.3) users; $t(53)=-0.5$, $p=0.618$, excitement felt by frequent (M=5.1, SD=1.7) and non-frequent (M=5.1, SD=1.4) users; $t(54)=0.04$, $p=0.967$, happiness felt by frequent (M=5.1, SD=1.6) and non-frequent (M=5.3, SD=1.2) users; $t(52)=-0.59$, $p=0.559$, satisfaction felt by frequent (M=5.3, SD=1.5) and non-frequent (M=5.3, SD=1.3) users; $t(58)=-0.07$, $p=0.947$, disappointment felt by frequent (M=2.9, SD=1.6) and non-frequent (M=3.2, SD=1.9) users; $t(76)=-0.79$, $p=0.43$, pressure felt by frequent (M=3.2, SD=1.9) and non-frequent (M=3, SD=1.9) users; $t(65)=0.45$, $p=0.652$, stress felt by frequent (M=3, SD=1.9) and non-frequent (M=2.9, SD=1.9) users; $t(65)=0.15$, $p=0.878$, irritation felt by frequent (M=2.9, SD=1.8) and non-frequent (M=2.8, SD=1.9) users; $t(66)=0.39$, $p=0.698$, and manipulation felt by frequent (M=3.8, SD=2) and non-frequent (M=3.5, SD=2.2) users; $t(70)=0.49$, $p=0.625$.

4.1.2 Dark Patterns Noticed. There were no significant differences in the results for nagging noticed by frequent (M=4.4, SD=1.7) and non-frequent (M=3.7, SD=1.9) users; $t(73)=1.82$, $p=0.073$, obstruction

noticed by frequent (M=4.3, SD=1.5) and non-frequent (M=3.6, SD=1.9) users; $t(78)=1.96$, $p=0.054$, sneaking noticed by frequent (M=4.2, SD=1.7) and non-frequent (M=3.6, SD=1.9) users; $t(72)=1.61$, $p=0.112$, interface interference noticed by frequent (M=4.1, SD=1.6) and non-frequent (M=3.6, SD=1.7) users; $t(71)=1.55$, $p=0.126$, and forced action noticed by frequent (M=3.7, SD=1.9) and non-frequent (M=3.3, SD=1.9) users; $t(66)=0.84$, $p=0.403$.

4.1.3 Correlation Between Emotions Felt and Dark Patterns Noticed. There was little to no significant correlations found between delight $r(98)=0.22$, excitement $r(98)=0.28$, happiness $r(98)=0.29$, satisfaction $r(98)=0.27$, disappointment $r(98)=0.53$, pressure $r(98)=0.56$, stress $r(98)=0.58$, irritation $r(98)=0.57$, and manipulation $r(98)=0.6$ and dark patterns noticed.

4.2 Amazon Results

4.2.2 Felt Emotions. There were no significant differences in the scores for delight felt by daily (M=5.1, SD=1.4) and non-daily (M=4.5, SD=1.6) users; $t(98)=1.85$, $p=0.067$, excitement felt by daily (M=5.3, SD=1.5) and non-daily (M=4.7, SD=1.7) users; $t(98)=1.74$, $p=0.086$, happiness felt by daily (M=5.4, SD=1.5) and non-daily (M=4.9, SD=1.6) users; $t(98)=1.64$, $p=0.105$, satisfaction felt by daily (M=5.6, SD=1.5) and non-daily (M=5.3, SD=1.6) users; $t(97)=1.07$, $p=0.287$, disappointment felt by daily (M=3.3, SD=1.8) and non-daily (M=2.6, SD=1.8) users; $t(97)=1.95$, $p=0.053$, pressure felt by daily (M=3.2, SD=2) and non-daily (M=2.5, SD=1.8) users; $t(95)=1.84$, $p=0.069$, and manipulation felt by daily (M=3.8, SD=2.1) and non-daily (M=3.3, SD=2.2) users; $t(97)=1.18$, $p=0.241$.

However, there was a significant difference in the scores for stress felt by daily (M=3, SD=1.9) and non-daily (M=2.3, SD=1.7) users; $t(92)=2.06$, $p=0.042$, and irritation felt by daily (M=3, SD=1.8) and non-daily (M=2.2, SD=1.6) users; $t(94)=2.4$, $p=0.018$.

4.2.2 Dark Patterns Noticed. There were no significant differences in the scores for nagging noticed by daily (M=4, SD=1.8) and non-daily (M=3.4, SD=1.9) users; $t(98)=1.63$, $p=0.106$.

However, there was a significant difference in the scores for obstruction noticed by daily (M=3.8, SD=1.8) and non-daily (M=3, SD=2) users; $t(98)=2.13$, $p=0.036$, sneaking noticed by daily (M=3.6, SD=1.6) and non-daily (M=2.8, SD=1.8) users; $t(98)=2.36$, $p=0.02$, interface interference noticed by daily (M=3.9, SD=1.8) and non-daily (M=3, SD=2) users; $t(98)=2.41$, $p=0.018$, and forced action noticed by daily (M=3.8, SD=1.8) and non-daily (M=2.7, SD=1.7) users; $t(96)=3.19$, $p=0.002$.

4.2.3 Correlation Between Emotions Felt and Dark Patterns Noticed. There was little to no significant correlations found between delight $r(100)=0.16$, excitement $r(100)=0.21$, happiness $r(100)=0.2$, satisfaction $r(100)=0.04$, disappointment $r(100)=0.61$, pressure $r(100)=0.64$, stress $r(100)=0.64$, irritation $r(100)=0.65$, and manipulation $r(100)=0.62$ and dark patterns noticed.

4.3 Wish Results

4.3.1 Felt Emotions. There were no significant differences in the scores for delight felt by daily (M=4.9, SD=1.6) and non-daily (M=4.5, SD=1.6) users; $t(50)=0.95$, $p=0.348$, excitement felt by daily (M=4.7, SD=1.7) and non-daily (M=4.9, SD=1.5) users; $t(45)=-0.64$, $p=0.524$, happiness felt by daily (M=5, SD=1.9) and non-daily (M=5, SD=1.6) users; $t(42)=-0.07$, $p=0.945$, satisfaction felt by daily (M=4.9, SD=2) and non-daily (M=5, SD=1.4) users; $t(38)=-0.37$, $p=0.713$, disappointment felt by daily (M=3.8, SD=2.1) and non-daily (M=3.3, SD=1.8) users; $t(44)=1.02$, $p=0.314$, pressure felt by daily (M=3.1, SD=1.9) and non-daily (M=3, SD=1.8) users; $t(46)=0.14$, $p=0.892$, stress felt by daily (M=3.1, SD=1.9)

and non-daily (M=3.1, SD=2) users; $t(50)=-0.03$, $p=0.978$, and irritation felt by daily (M=3, SD=1.8) and non-daily (M=3.1, SD=1.9) users; $t(53)=-0.18$, $p=0.854$.

However, there was a significant difference in the scores for manipulation felt by daily (M=4.8, SD=1.9) and non-daily (M=3.8, SD=2.1) users; $t(54)=2.29$, $p=0.026$.

4.3.2 Dark Patterns Noticed. There were no significant differences in the scores for nagging noticed by daily (M=4.3, SD=2) and non-daily (M=4, SD=1.9) users; $t(48)=0.76$, $p=0.449$, obstruction noticed by daily (M=4, SD=1.8) and non-daily (M=3.9, SD=1.8) users; $t(50)=0.34$, $p=0.732$, sneaking noticed by daily (M=4.1, SD=1.9) and non-daily (M=3.9, SD=1.8) users; $t(47)=0.49$, $p=0.629$, interface interfering noticed by daily (M=4.1, SD=1.9) and non-daily (M=3.9, SD=1.7) users; $t(45)=0.43$, $p=0.666$, and forced action noticed by daily (M=4.2, SD=2) and non-daily (M=3.7, SD=1.9) users; $t(47)=1.18$, $p=0.243$.

4.3.3 Correlation Between Emotions Felt and Dark Patterns Noticed. There was little to no significant correlations found between delight $r(100)=0.26$, excitement $r(100)=0.18$, happiness $r(100)=0.25$, satisfaction $r(100)=0.26$, disappointment $r(100)=0.43$, pressure $r(100)=0.5$, stress $r(100)=0.55$, irritation $r(100)=0.57$, and manipulation $r(100)=0.56$ and dark patterns noticed.

4.4 PUBG Mobile Results

4.4.1 Felt Emotions. There were no significant differences in the scores for delight felt by daily (M=4.8, SD=1.5) and non-daily (M=4.8, SD=1.4) users; $t(97)=0.32$, $p=0.746$, excitement felt by daily (M=5.1, SD=1.7) and non-daily (M=5.1, SD=1.4) users; $t(99)=0.08$, $p=0.939$, happiness felt by daily (M=5.4, SD=1.5) and non-daily (M=5.4, SD=1.5) users; $t(90)=0.21$, $p=0.837$, satisfaction felt by daily (M=5.3, SD=1.6) and non-daily (M=5, SD=1.3) users; $t(99)=0.82$, $p=0.416$, disappointment felt by daily (M=3.1, SD=1.7) and non-daily (M=3.2, SD=1.8) users; $t(89)=-0.27$, $p=0.79$, pressure felt by daily (M=3.4, SD=1.8) and non-daily (M=3.5, SD=1.7) users; $t(96)=-0.53$, $p=0.598$, stress felt by daily (M=3.4, SD=1.9) and non-daily (M=3.8, SD=1.8) users; $t(94)=-0.95$, $p=0.344$, irritation felt by daily (M=3.4, SD=1.9) and non-daily (M=3.3, SD=1.6) users; $t(98)=0.42$, $p=0.672$, and manipulation felt by daily (M=4.1, SD=1.9) and non-daily (M=3.6, SD=2.2) users; $t(83)=1.21$, $p=0.23$.

4.4.2 Game Mechanics Noticed. There were no significant differences in the scores for grinding noticed by daily (M=4.5, SD=1.8) and non-daily (M=4.4, SD=1.6) users; $t(97)=0.25$, $p=0.802$, playing by appointment noticed by daily (M=4.3, SD=1.8) and non-daily (M=4.1, SD=2) users; $t(89)=0.62$, $p=0.535$, pay to skip noticed by daily (M=4.2, SD=2) and non-daily (M=3.8, SD=2.1) users; $t(90)=0.89$, $p=0.377$, pay wall noticed by daily (M=4.3, SD=1.8) and non-daily (M=3.6, SD=1.9) users; $t(92)=1.86$, $p=0.066$, pay to win noticed by daily (M=4.5, SD=1.8) and non-daily (M=4.1, SD=1.7) users; $t(95)=1.14$, $p=0.256$, social pyramid schemes noticed by daily (M=4, SD=1.9) and non-daily (M=3.9, SD=1.9) users; $t(92)=0.4$, $p=0.686$, impersonation noticed by daily (M=4.3, SD=1.7) and non-daily (M=3.7, SD=1.9) users; $t(88)=1.63$, $p=0.106$, infinite treadmill noticed by daily (M=4, SD=1.8) and non-daily (M=4, SD=1.9) users; $t(91)=0$, $p=1$, and power creep noticed by daily (M=4.4, SD=1.7) and non-daily (M=4, SD=1.6) users; $t(95)=1.32$, $p=0.189$.

4.4.3 Correlation Between Emotions Felt and Game Mechanics Noticed. There was little to no significant correlations found between delight $r(102)=0.21$, excitement $r(102)=0.29$, happiness $r(102)=0.33$, satisfaction $r(102)=0.28$, disappointment $r(102)=0.5$, pressure $r(102)=0.47$, stress $r(102)=0.39$, irritation $r(102)=0.42$, and manipulation $r(102)=0.55$ and game mechanics noticed.

4.5 Pokémon Go Results

4.5.1 Emotions Felt. There were no significant differences in the scores for delight felt by daily (M=4.9, SD=1.6) and non-daily (M=4.7, SD=1.5) users; $t(91)=0.71$, $p=0.482$, excitement felt by daily (M=5.2, SD=1.8) and non-daily (M=5.1, SD=1.5) users; $t(85)=0.32$, $p=0.747$, happiness felt by daily (M=5.3, SD=1.7) and non-daily (M=5.2, SD=1.4) users; $t(86)=0.24$, $p=0.81$, satisfaction felt by daily (M=5.2, SD=1.7) and non-daily (M=5, SD=1.3) users; $t(81)=0.79$, $p=0.429$, disappointment felt by daily (M=3.1, SD=1.8) and non-daily (M=2.7, SD=1.7) users; $t(93)=1.14$, $p=0.256$, pressure felt by daily (M=3.1, SD=1.7) and non-daily (M=2.6, SD=1.8) users; $t(96)=1.38$, $p=0.171$, stress felt by daily (M=2.9, SD=1.7) and non-daily (M=2.7, SD=1.8) users; $t(96)=0.56$, $p=0.576$, irritation felt by daily (M=2.6, SD=1.5) and non-daily (M=2.5, SD=1.7) users; $t(98)=0.16$, $p=0.875$, and manipulation felt by daily (M=3.3, SD=2.1) and non-daily (M=2.8, SD=2) users; $t(92)=1.13$, $p=0.259$.

4.5.2 Game Mechanics Noticed. There were no significant differences in the scores for grinding noticed by daily (M=4.7, SD=1.6) and non-daily (M=4.1, SD=1.8) users; $t(97)=1.68$, $p=0.097$, playing by appointment noticed by daily (M=4.1, SD=2.1) and non-daily (M=4, SD=2) users; $t(94)=0.26$, $p=0.795$, pay to skip noticed by daily (M=3.8, SD=2.1) and non-daily (M=3.5, SD=2.1) users; $t(94)=0.69$, $p=0.492$, pay wall noticed by daily (M=4, SD=2) and non-daily (M=3.3, SD=2) users; $t(95)=1.71$, $p=0.091$, pay to win noticed by daily (M=4.3, SD=2) and non-daily (M=4.1, SD=2.1) users; $t(96)=0.44$, $p=0.66$, social pyramid schemes noticed by daily (M=3.8, SD=2) and non-daily (M=3.1, SD=2) users; $t(94)=1.84$, $p=0.069$, impersonation noticed by daily (M=3.6, SD=2) and non-daily (M=3.1, SD=2) users; $t(95)=1.13$, $p=0.261$, infinite treadmill noticed by daily (M=4, SD=2) and non-daily (M=3.6, SD=2.1) users; $t(96)=1.09$, $p=0.279$, and power creep noticed by daily (M=4.2, SD=1.8) and non-daily (M=3.6, SD=2) users; $t(97)=1.58$, $p=0.118$.

4.5.3 Correlation Between Emotions Felt and Game Mechanics Noticed. There was little to no significant correlations found between delight $r(100)=0.14$, excitement $r(100)=0.19$, happiness $r(100)=0.16$, satisfaction $r(100)=0.12$, disappointment $r(100)=0.37$, pressure $r(100)=0.39$, stress $r(100)=0.41$, irritation $r(100)=0.42$, and manipulation $r(100)=0.47$ and game mechanics noticed.

4.6 Subway Surfer Results

4.6.1 Emotions Felt. There were no significant differences in the scores for delight felt by daily (M=5.1, SD=1.6) and non-daily (M=4.5, SD=1.6) users; $t(65)=1.81$, $p=0.075$, excitement felt by daily (M=5.2, SD=1.8) and non-daily (M=4.7, SD=1.6) users; $t(58)=1.18$, $p=0.242$, happiness felt by daily (M=5.3, SD=1.6) and non-daily (M=5.2, SD=1.6) users; $t(62)=0.23$, $p=0.817$, satisfaction felt by daily (M=5.1, SD=1.7) and non-daily (M=5.1, SD=1.5) users; $t(57)=0.17$, $p=0.864$, and pressure felt by daily (M=3.4, SD=2.2) and non-daily (M=2.6, SD=1.8) users; $t(53)=1.83$, $p=0.073$.

However, there was a significant difference in the scores for disappointment felt by daily (M=3.7, SD=2.1) and non-daily (M=2.4, SD=1.6) users; $t(52)=3.06$, $p=0.004$, stress felt by daily (M=3.4, SD=2) and non-daily (M=2.3, SD=1.5) users; $t(49)=2.71$, $p=0.009$, irritation felt by daily (M=3.4, SD=2) and non-daily (M=2.3, SD=1.5) users; $t(49)=2.75$, $p=0.008$, and manipulation felt by daily (M=4.1, SD=2.1) and non-daily (M=2.8, SD=2.2) users; $t(66)=2.94$, $p=0.004$.

4.6.2 Game Mechanics Noticed. There were no significant differences in the scores for pay to win noticed by daily (M=4.4, SD=1.9) and non-daily (M=3.7, SD=1.9) users; $t(62)=1.7$, $p=0.094$, social pyramid schemes noticed by daily (M=4.3, SD=2) and non-daily (M=3.3, SD=1.9) users; $t(62)=2.48$, $p=0.016$, and infinite treadmill noticed by daily (M=4.1, SD=2) and non-daily (M=3.6, SD=2) users; $t(64)=1.21$, $p=0.229$.

However, there was a significant difference in the scores for grinding noticed by daily (M=4.7, SD=1.7) and non-daily (M=3.8, SD=1.8) users; $t(67)=2.39$, $p=0.02$, playing by appointment noticed by daily (M=4.2, SD=1.9) and non-daily (M=3.3, SD=1.6) users; $t(54)=2.41$, $p=0.019$, pay to skip noticed by daily (M=4.5, SD=1.8) and non-daily (M=3.5, SD=1.8) users; $t(65)=2.77$, $p=0.007$, pay wall noticed by daily

($M=4.4$, $SD=1.7$) and non-daily ($M=3.1$, $SD=1.7$) users; $t(62)=3.61$, $p=0.001$, impersonation noticed by daily ($M=4$, $SD=2.1$) and non-daily ($M=3$, $SD=1.8$) users; $t(57)=2.3$, $p=0.025$, and power creep noticed by daily ($M=4.2$, $SD=1.9$) and non-daily ($M=3$, $SD=1.8$) users; $t(61)=3.03$, $p=0.004$.

4.6.3 Correlation Between Emotions Felt and Game Mechanics Noticed. There was little to no significant correlations found between delight $r(100)=0.26$, excitement $r(100)=0.36$, happiness $r(100)=0.31$, satisfaction $r(100)=0.29$, disappointment $r(100)=0.5$, pressure $r(100)=0.52$, stress $r(100)=0.51$, irritation $r(100)=0.53$, and manipulation $r(100)=0.58$ and game mechanics noticed.

4.7 Summary of Results and Tables

4.7.1 Emotions Felt Summary.

Table #. Summary of mean ($\bar{x}$), variance (σ^2), degrees of freedom (df), t-statistic (t-stat), and p-value (p) for emotions felt by frequent and non-frequent users.

	Frequent		Non-Frequent				
	$\bar{x}$	σ^2	$\bar{x}$	σ^2	df	t-stat	p
AliExpress							
Delighted	4.67	2.67	4.83	1.71	53	-0.50	0.618
Excited	5.09	2.84	5.08	1.85	54	0.04	0.967
Happy	5.12	2.55	5.31	1.53	52	-0.59	0.559
Satisfied	5.30	2.16	5.32	1.72	58	-0.07	0.947
Disappointed	2.88	2.55	3.17	3.71	76	-0.79	0.430
Pressured	3.15	3.51	2.97	3.59	65	0.45	0.652
Stressed	2.97	3.53	2.91	3.62	65	0.15	0.878
Irritable	2.94	3.37	2.78	3.58	66	0.39	0.698
Manipulated	3.76	4.06	3.54	4.91	70	0.49	0.625
Amazon							
Delighted	5.09	1.95	4.53	2.60	98	1.85	0.067
Excited	5.26	2.37	4.70	2.79	98	1.74	0.086
Happy	5.43	2.12	4.92	2.57	98	1.64	0.105
Satisfied	5.64	2.37	5.30	2.56	97	1.07	0.287
Disappointed	3.30	3.08	2.60	3.21	97	1.95	0.053
Pressured	3.21	3.87	2.51	3.37	95	1.84	0.069
Stressed	3.02	3.76	2.26	2.89	92	2.06	0.042
Irritable	3.00	3.22	2.17	2.72	94	2.40	0.018
Manipulated	3.83	4.49	3.32	4.80	97	1.18	0.241
Wish							
Delighted	4.86	2.42	4.53	2.48	50	0.95	0.348
Excited	4.71	2.73	4.94	2.22	45	-0.64	0.524
Happy	5.00	3.56	5.03	2.45	42	-0.07	0.945
Satisfied	4.86	4.20	5.01	2.07	38	-0.37	0.713
Disappointed	3.79	4.25	3.33	3.27	44	1.02	0.314
Pressured	3.07	3.70	3.01	3.17	46	0.14	0.892

Stressed	3.07	3.70	3.08	3.88	50	-0.03	0.978
Irritable	3.04	3.22	3.11	3.71	53	-0.18	0.854
Manipulated	4.82	3.63	3.82	4.43	54	2.29	0.026
PUBG Mobile							
Delighted	4.84	2.38	4.75	1.96	97	0.32	0.746
Excited	5.14	3.03	5.11	2.06	99	0.08	0.939
Happy	5.45	2.11	5.39	2.38	90	0.21	0.837
Satisfied	5.26	2.48	5.02	1.79	99	0.82	0.416
Disappointed	3.09	2.92	3.18	3.41	89	-0.27	0.790
Pressured	3.36	3.25	3.55	2.81	96	-0.53	0.598
Stressed	3.40	3.58	3.75	3.35	94	-0.95	0.344
Irritable	3.40	3.47	3.25	2.61	98	0.42	0.672
Manipulated	4.07	3.43	3.57	4.95	83	1.21	0.230
Pokémon Go							
Delighted	4.89	2.46	4.67	2.15	91	0.71	0.482
Excited	5.20	3.30	5.09	2.23	85	0.32	0.747
Happy	5.31	2.76	5.24	1.92	86	0.24	0.810
Satisfied	5.24	2.87	5.00	1.70	81	0.79	0.429
Disappointed	3.11	3.15	2.71	2.95	93	1.14	0.256
Pressured	3.09	2.95	2.60	3.32	96	1.38	0.171
Stressed	2.87	2.80	2.67	3.15	96	0.56	0.576
Irritable	2.58	2.20	2.53	2.92	98	0.16	0.875
Manipulated	3.29	4.44	2.82	4.04	92	1.13	0.259
Subway Surfer							
Delighted	5.09	2.52	4.48	2.59	65	1.81	0.075
Excited	5.18	3.22	4.75	2.56	58	1.18	0.242
Happy	5.27	2.58	5.19	2.43	62	0.23	0.817
Satisfied	5.12	3.05	5.06	2.36	57	0.17	0.864
Disappointed	3.70	4.34	2.43	2.64	52	3.06	0.004
Pressured	3.39	4.75	2.60	3.09	53	1.83	0.073
Stressed	3.36	4.18	2.28	2.15	49	2.71	0.009
Irritable	3.39	4.18	2.30	2.15	49	2.75	0.008
Manipulated	4.12	4.30	2.81	4.64	66	2.94	0.004

4.7.2 Dark Patterns/Game Mechanics Noticed Summary.

Table #. Summary of mean ($\bar{x}$), variance (σ^2), degrees of freedom (df), t-statistic (t-stat), and p-value (p) for dark patterns noticed by frequent and non-frequent users.

	Frequent		Non-Frequent		df	t-stat	p
	$\bar{x}$	σ^2	$\bar{x}$	σ^2			
AliExpress							
Nagging	4.39	2.81	3.71	3.74	73	1.82	0.073

Obstruction	4.27	2.27	3.58	3.56	78	1.96	0.054
Sneaking	4.18	2.90	3.57	3.72	72	1.61	0.112
Interface Interference	4.12	2.42	3.58	3.03	71	1.55	0.126
Forced Action	3.67	3.60	3.32	3.75	66	0.84	0.403
Amazon							
Nagging	4.00	3.22	3.40	3.63	98	1.63	0.106
Obstruction	3.81	3.25	3.00	4.00	98	2.13	0.036
Sneaking	3.57	2.68	2.75	3.34	98	2.36	0.020
Interface Interference	3.94	3.41	3.02	3.87	98	2.41	0.018
Forced Action	3.79	3.17	2.66	3.04	96	3.19	0.002
Wish							
Nagging	4.32	3.93	3.99	3.76	48	0.76	0.449
Obstruction	4.00	3.26	3.86	3.30	50	0.34	0.732
Sneaking	4.14	3.46	3.94	3.07	47	0.49	0.629
Interface Interference	4.07	3.77	3.89	3.00	45	0.43	0.666
Forced Action	4.21	4.03	3.69	3.54	47	1.18	0.243
PUBG Mobile							
Grinding	4.52	3.20	4.43	2.67	97	0.25	0.802
Playing by Appointment	4.33	3.31	4.09	3.85	89	0.62	0.535
Pay to Skip	4.16	3.85	3.80	4.31	90	0.89	0.377
Pay Wall	4.33	3.42	3.64	3.49	92	1.86	0.066
Pay to Win	4.50	3.41	4.09	3.06	95	1.14	0.256
Social Pyramid Schemes	4.02	3.53	3.86	3.66	92	0.40	0.686
Impersonation	4.28	2.98	3.68	3.57	88	1.63	0.106
Infinite Treadmill	4.00	3.30	4.00	3.49	91	0.00	1.000
Power Creep	4.41	2.91	3.98	2.58	95	1.32	0.189
Pokémon Go							
Grinding	4.69	2.72	4.11	3.25	97	1.68	0.097
Playing by Appointment	4.09	4.22	3.98	4.09	94	0.26	0.795
Pay to Skip	3.80	4.44	3.51	4.37	94	0.69	0.492
Pay Wall	3.96	3.91	3.27	4.02	95	1.71	0.091
Pay to Win	4.31	4.04	4.13	4.59	96	0.44	0.660
Social Pyramid Schemes	3.78	3.81	3.05	3.83	94	1.84	0.069
Impersonation	3.60	3.93	3.15	4.09	95	1.13	0.261
Infinite Treadmill	4.04	3.86	3.60	4.43	96	1.09	0.279
Power Creep	4.16	3.13	3.56	3.92	97	1.58	0.118
Subway Surfer							
Grinding	4.70	2.97	3.81	3.31	67	2.39	0.020
Playing by Appointment	4.21	3.55	3.30	2.42	54	2.41	0.019
Pay to Skip	4.55	3.13	3.49	3.31	65	2.77	0.007
Pay Wall	4.36	2.93	3.06	2.78	62	3.61	0.001
Pay to Win	4.42	3.75	3.73	3.47	62	1.70	0.094
Social Pyramid Schemes	4.27	3.83	3.25	3.56	62	2.48	0.016

Impersonation	4.00	4.25	3.03	3.27	57	2.30	0.025
Infinite Treadmill	4.12	3.86	3.61	3.94	64	1.21	0.229
Power Creep	4.24	3.56	3.04	3.26	61	3.03	0.004

4.7.3 Correlation Between Emotions Felt and Game Mechanics Noticed Summary.

Table #. Correlation Results Between Dark Patterns Noticed and Emotions Felt by All Participants of Each Survey

	Delight	Excitement	Happiness	Satisfaction	Disappointment	Pressure	Stress	Irritation	Manipulation
AliExpress									
Nagging	0.22	0.29	0.32	0.28	0.47	0.52	0.49	0.50	0.59
Obstruction	0.22	0.28	0.22	0.18	0.51	0.53	0.53	0.50	0.55
Sneaking	0.26	0.29	0.32	0.31	0.52	0.54	0.60	0.57	0.63
Interface Interference	0.25	0.34	0.32	0.34	0.54	0.53	0.63	0.61	0.54
Forced Action	0.13	0.21	0.26	0.23	0.62	0.67	0.66	0.66	0.70
Amazon									
Nagging	0.13	0.13	0.15	0.04	0.56	0.57	0.62	0.61	0.58
Obstruction	0.15	0.20	0.17	0.04	0.63	0.59	0.62	0.60	0.65
Sneaking	0.23	0.27	0.28	0.08	0.64	0.69	0.66	0.65	0.61
Interface Interference	0.14	0.19	0.22	0.11	0.59	0.62	0.62	0.67	0.63
Forced Action	0.15	0.24	0.16	0.04	0.64	0.72	0.67	0.71	0.65
Wish									
Nagging	0.30	0.11	0.20	0.24	0.43	0.53	0.54	0.61	0.59
Obstruction	0.29	0.25	0.34	0.31	0.40	0.52	0.55	0.52	0.60
Sneaking	0.31	0.21	0.33	0.33	0.39	0.44	0.48	0.51	0.50
Interface Interference	0.26	0.21	0.23	0.24	0.34	0.40	0.52	0.49	0.49
Forced Action	0.16	0.11	0.16	0.16	0.56	0.61	0.66	0.70	0.62
PUBG Mobile									
Grinding	0.35	0.38	0.40	0.31	0.42	0.38	0.30	0.39	0.35
Playing by Appointment	0.19	0.37	0.43	0.29	0.43	0.44	0.36	0.34	0.58
Pay to Skip	0.27	0.25	0.35	0.28	0.51	0.48	0.41	0.43	0.56
Pay Wall	0.18	0.27	0.33	0.34	0.50	0.49	0.43	0.44	0.59
Pay to Win	0.19	0.31	0.31	0.27	0.50	0.39	0.35	0.47	0.48
Social Pyramid Schemes	0.19	0.28	0.33	0.30	0.54	0.51	0.50	0.39	0.59
Impersonation	0.21	0.27	0.28	0.32	0.54	0.51	0.42	0.53	0.55
Infinite Treadmill	0.14	0.21	0.28	0.21	0.55	0.49	0.40	0.37	0.58
Power Creep	0.16	0.23	0.28	0.20	0.52	0.58	0.36	0.41	0.65

Pokémon Go

Grinding	0.35	0.20	0.19	0.14	0.27	0.21	0.32	0.28	0.28
Playing by Appointment	0.07	0.18	0.06	0.03	0.42	0.37	0.43	0.45	0.42
Pay to Skip	0.12	0.15	0.13	0.06	0.47	0.49	0.56	0.51	0.55
Pay Wall	0.02	0.14	0.04	0.05	0.44	0.40	0.49	0.52	0.56
Pay to Win	0.11	0.22	0.21	0.10	0.32	0.35	0.38	0.30	0.37
Social Pyramid Schemes	0.16	0.20	0.19	0.16	0.37	0.49	0.42	0.49	0.59
Impersonation	0.14	0.21	0.23	0.21	0.37	0.50	0.40	0.45	0.57
Infinite Treadmill	0.18	0.25	0.27	0.22	0.32	0.31	0.27	0.38	0.37
Power Creep	0.12	0.19	0.13	0.11	0.38	0.37	0.39	0.39	0.48

Subway Surfer

Grinding	0.31	0.33	0.32	0.25	0.38	0.45	0.42	0.44	0.44
Playing by Appointment	0.37	0.47	0.34	0.34	0.59	0.59	0.63	0.64	0.63
Pay to Skip	0.26	0.38	0.29	0.31	0.45	0.46	0.46	0.45	0.50
Pay Wall	0.25	0.45	0.32	0.33	0.59	0.57	0.57	0.60	0.63
Pay to Win	0.06	0.23	0.25	0.25	0.37	0.45	0.41	0.43	0.54
Social Pyramid Schemes	0.24	0.35	0.30	0.27	0.49	0.54	0.52	0.53	0.63
Impersonation	0.40	0.41	0.41	0.33	0.52	0.56	0.49	0.55	0.69
Infinite Treadmill	0.17	0.24	0.25	0.30	0.50	0.47	0.44	0.48	0.46
Power Creep	0.29	0.33	0.29	0.26	0.59	0.61	0.61	0.61	0.67

6 Discussion

We delved into the various factors affecting the efficacy of dark patterns by analyzing frequent and non-frequent users of certain applications (AliExpress, Amazon, Wish, PUBG Mobile, Pokémon Go, and Subway Surfer). We sought to determine if there was a relationship between frequency of use and a user's awareness of dark patterns on said application. To do this, we developed a survey to target daily users of these applications and asked them to share their experiences on these sites. We analyzed the data through descriptive and statistical analysis.

While we did not find many statistically significant differences in means for frequency of use and emotions felt or dark patterns/game mechanics noticed, we did provide a basis for future research to delve deeper into this area. For instance, the means for four out of five dark patterns noticed by frequent and non-frequent users of Amazon were statistically significant. Frequent users of Amazon consistently ranked that they noticed obstruction, sneaking, interface interference, and forced action higher compared to their non-frequent counterparts. This trend was not observed by any of the other e-commerce sites.

One note of importance is the difference in the face-value experiences on all three sites. Clicking on the Google link for wish.com leads directly to a login page which requires users to sign up. However, there is a way to browse Wish without signing up by accessing their help page labeled, 'How do I browse Wish without signing in?' Clicking on the Google link for aliexpress.com leads straight to the home page of AliExpress. However, before users can browse the merchandise, two pop-up notifications appear. One asks the user to subscribe to notifications (utilizing a bright red button labelled 'Allow' and a slightly grey button for 'Don't Allow') and the other informs the users that they have received a coupon. On the other hand, clicking on the Google link for amazon.ca leads the user straight to the home page. A small pop-up appears by the 'Hello, Sign in' nav bar option that prompts the user to sign in but it disappears after a while. Users

can start browsing immediately on Amazon. All three sites provide some proof of utilizing dark pattern techniques at face-value, but some are more covert than others. This could lead to future work in determining a framework to analyze the coyness of dark patterns in specific applications.

6.# Persuasion Knowledge Theory and Manipulation Literacy

Our data has suggested that awareness of dark patterns or game mechanics prior to taking the survey may not affect whether users continue to be daily users or not. Due to the fact that the percentages of users aware vs not aware for daily and non-daily users are similar or tend to be more aware if as a daily user, this may well indicate that users do not use their potential knowledge of manipulative techniques as we have previously thought.

Friestad and Wright developed their Persuasion Knowledge Model (PKM) in 1994 when they proposed the idea that consumer's prior knowledge of persuasion techniques enabled them to cope (response to a perceived persuasion attempt) and protect themselves against such strategies. However, a vital element of the PKM posits that awareness of persuasive techniques being used is key to the consumer's ability to access their persuasion knowledge. Without awareness, PKM theorizes that consumers cannot access their prior persuasive knowledge to protect themselves [24].

Lewis came to the same conclusion in 2013, proposing the idea of manipulation literacy, "the ability for users to identify manipulative techniques, understand their possible effects, and then make consensual, informed choices about how to proceed. [19]" Lewis stated the importance of awareness of dark patterns, suggesting that dark patterns lost their effectiveness once a user was aware of them. Lewis also provided an example which demonstrated that knowledge gained from increased manipulation literacy enabled users to better protect themselves from potential manipulation techniques.

The PKM and Manipulation literacy theory both suggest that users may develop negative feelings (ex. skepticism) towards the usage of such manipulative techniques after recognition of such techniques.

However, our data suggest that daily users of e-commerce websites had similar percentages of those with prior knowledge of dark patterns prior to the survey compared with those who were not daily users, implying that prior knowledge of dark patterns did not deter daily users much. Daily users in the mobile applications had higher or similar knowledge of game mechanics prior to the survey compared to the non-daily users. While it cannot be confirmed that these game mechanics are inherently dark, there was proof in the qualitative responses of the surveys that indicated that participants felt some sense of manipulative techniques being employed within the game. The percentage of daily users who knew about the game mechanics indicates that there may be more to the assumption that increased knowledge on dark patterns leads to negative feelings towards the pattern. These findings are in conjunction with Evans & Park who also provided evidence that there are alternatives to the base assumptions made by PKM [25].

6.# The Darkness in Dark Game Design Patterns and the Unfair vs Unethical Debate

It's been thoroughly agreed upon that the concepts stated by Gray et al. are definitely dark patterns. However, the same cannot be stated for the many game mechanics seen in the three applications we analyzed. Stated by Zagal et al. and also elaborated on by Lewis, deciding whether something is a dark pattern in a game is very difficult. It has been proposed that (find other examples of dark patterns taxonomy but not from dark pattern literature).

There is much debate about what constitutes as a dark pattern within game design. Although, Zagal et al. provided a taxonomy with three overarching themes (monetary, social-capital, and temporal) and examples of dark game design patterns that fell under each category, there is still no concrete way to determine what

constitutes a game mechanic or feature as dark. Infact, Deterding et al. illustrates the lack of any empirical evidence to support the example dark patterns in the Dark Patterns in the Design of Games paper in support of their three-part argument why the proposal of dark game design patterns are counterproductive to the ethical debate of darkness in game design.

Lewis proposed additional dark patterns found in games in his Motivational Design Patterns book. While he provided criteria for what constituted as dark for grinding, he failed to provide such specific criteria for any of the other dark patterns listed. For example, monetized rivalries (i.e. pay to win) is described as “Exploiting user competitiveness to incentivize purchasing.” While he gave a concrete example of when monetized rivalries happen, there are many other ways that it can happen. Since his dissertation was published in 2013, mobile games have changed drastically along with the concept of what constitutes as pay to win as more mobile games find different ways to monetize their apps (citations?).

Karhulahti and Kimppa (2018) noted that pay-to-play game play enabled an unfair advantage for those that paid over those that didn’t [28]. This unfair advantage was also noted by a commentator in Lewis (2013) paper which noted the unfairness when starting a game now versus many years ago [19]. For example, this would be like someone paying for world class training in soccer and then playing a pick-up game with a bunch of amateurs who also had started training on their own at the same time. The one who bought professional training would have an unfair advantage over the amateurs. As noted by Hyrnsalmi & Kimppa (2020), the mobile platform has transformed apps from a product into a service, monetizing the business model of mobile development [27].

7 Limitations

Some limitations include sample bias in this study. Our main method of acquiring participants was through Amazon Mechanical Turk, a website that pays users to take surveys. However, our preview survey participants (which tested Pokémon Go and Amazon) were taken from various social media sites and forums designed specifically for Pokémon Go. We found statistically significant differences between the noticed game mechanics in the Pokémon Go preview survey and the Amazon Mechanical Turk survey. These discrepancies suggest that one or the other (or both) surveys are not decent representations of the population as a whole. However, this does not reduce their effectiveness in being representations of their own specific subpopulations.

Another limitation was low n in the Amazon Mechanical Turk surveys that we published. We only found weak evidence that there might be a relationship between time spent in an application and the amount of dark patterns noticed (in Amazon survey). A higher n would further cement the idea that the hypothesis is correct or wrong.

8 Future Work

There is still a multitude of factors affecting the efficacy of dark patterns that have yet to be analyzed. Future work can continue to build on the growing base of literature that has focused on these factors. As mentioned in Section 2, work has already been done to document the effects of culture, education, and levels of aggressiveness in implementing dark patterns. This paper has touched upon frequency of use and prior awareness of dark patterns. As one example, future work could delve into how covert dark patterns affect awareness.

Another area of future work is determining the criteria for understanding what makes a game mechanic dark. While Zagal et al. (2013) provided three overarching themes (temporal, monetary, and social-capital dark patterns) under which dark game design patterns could fall under, there is still very limited criteria for which future researchers can determine what constitutes as dark. Something similar to Lewis’ criteria for grinding would strengthen the taxonomy by Zagal et al. (2013).

Furthermore, future research can be done to cement the idea of the PKM for covert advertising. When it was initially proposed, there were no such things as interface interference as we know it today. The PKM

was proposed for advertising in 1994. However, as advertising and digital persuasion becomes more covert and subtle, more research into how this affects the model should be done to either confirm that it is still relevant or if it should be updated.

As expressed by many others, coping techniques for dark patterns are vital. It has already been proposed that raising awareness is a good tactic to combat these deceptive online practices. However, as more research appears in the topic of dark patterns, we may need to take it a step further.

9 Conclusion

In this paper, we summarized the findings of other works of literature that have analyzed factors affecting the efficacy of dark patterns. We found that there were many factors to still be studied and decided to focus on frequency of use. We hypothesized that frequent users of applications would notice dark patterns more often than non-frequent users. However, our results, contrary to our hypothesis, reported that frequent and non-frequent users could notice these dark patterns equally well in most cases. We also provided a descriptive analysis of our data in relation to awareness of participants. Future work can build off this paper by determining other factors that affect people's susceptibility to dark patterns or by delving into the covertness of dark patterns and how they affect the awareness of users.

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